

The Coherence-Delay Trade-off in Tracking Under Drift

A Cube-Root Law for Finite-Memory Estimation

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Abstract

Tracking in a changing environment is governed by a simple tension: more memory reduces noise, but it also makes the estimate older. In passive streams this is already a statistical problem; in closed-loop systems it becomes an epistemic one, because the same actions used to stabilize behavior can also make a stale model look correct. The result is a two-part problem: a finite-memory tracking limit, and a masking diagnostic for systems whose actions shape what they later observe.

The first contribution formalizes a coherence-delay trade-off for finite-memory tracking under drift. For the finite-memory averaging class studied here, variance falls with memory while lag grows with drift and effective age, yielding a **cube-root law**: the **minimum achievable tracking error** scales as $\zeta^{1/3}$ with drift rate ζ , and the **optimal effective memory** scales as $\zeta^{-2/3}$. Empirically, sliding windows and EWMA recover slopes of 0.316 and 0.317, closely matching the theoretical prediction.

The second contribution concerns the regime in which low observed error is maintained not by adaptation but by steering the environment toward a stale model. We call this *coercive masking* and introduce the **Coherence Index (CI)** and the **Effort-Adjusted Coherence Index (CI^E)** to separate genuine adaptation from action-induced alignment. On passive benchmarks, generic changepoint detectors remain the stronger default alarms. On active synthetic and logged recommendation benchmarks, however, CI^E exposes masking that passive detectors do not encode under matched calibration, identifying masking onset with median delay 5.0 steps in the particle benchmark and improving KuaiRand bubble detection from 0.458 to 0.692 on a proxy-level logged benchmark, with paired user-level bootstrap intervals excluding zero.

The picture is straightforward. Under drift, finite memory imposes a real error floor. Under feedback, raw fit can be misleading unless the cost of intervention is made explicit. The theory explains the floor; the diagnostic reveals when apparent coherence is being purchased rather than earned.

Keywords: tracking under drift; finite-memory estimation; Wasserstein distance;

Sinkhorn divergence; concept drift; online compression; Coherence Index; Effort-Adjusted Coherence Index; coercive masking.
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1 Introduction

Modern adaptive systems do not observe a fixed world. They estimate, predict, rank, and control while the environment keeps changing beneath them, and in many cases their own actions help shape what they will observe next. In recommender systems, adaptive control, and streaming prediction, tracking under drift is not just a filtering problem. It is the question of how much of the past can still be trusted before it turns stale.

From that setting follow two distinct difficulties. The first is statistical. A finite-memory estimator improves when it can average more data, but every additional observation also comes from a more distant past. Under drift, memory reduces variance and increases staleness at the same time. The second is epistemic. In action-coupled systems, low observed error need not mean the model is current, because the system may be pushing the environment toward the model rather than updating the model to match the environment. A tracker can look coherent for the wrong reason.

These two difficulties belong to the same story. One asks what finite memory can achieve when the target itself is moving. The other asks how to diagnose the special failure mode in which apparent coherence is being maintained by intervention rather than by adaptation. The first yields a finite-memory law; the second yields a diagnostic for coercive masking.

The first result is a structural coherence-delay trade-off. When observations are aggregated over time, total tracking error decomposes into a variance term that falls with memory length and a lag term that rises with drift and the effective age of the data. Optimizing that balance yields a cube-root law for the finite-memory averaging class analyzed here: $\mathcal{E}_{\min} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$ with $n^* = (C_K/\zeta)^{2/3}$. This law is not presented as a universal theorem for all adaptive estimators. It is exact for the averaging class studied here, and in the critical-window regime the same exponent reappears as a minimax lower bound for arbitrary estimators restricted to the last m samples of a ζ -Lipschitz drift path. In that sense, tracking under drift is limited by the same object that makes online compression useful: memory itself.

The second result concerns feedback masking. In closed-loop settings, good fit can be purchased by control effort. The system may appear well aligned with its environment because its actions are holding the environment near a stale model. In the synthetic control regime, we call this *coercive masking*. To analyze it, we separate three functional roles: a representation of the world ($\mathcal{P}$), an actuation rule ($\mathcal{A}$), and an adaptation operator (Φ). This is a modeling decomposition for action-effective tracking systems, not a claim about every information process.

That decomposition yields two diagnostics: the Coherence Index (CI) and the Effort-Adjusted Coherence Index (CI^E). When explicit functional notation is needed, we write $\mathcal{C}(I(S, t), t)$ and $\mathcal{C}^E(I(S, t), t)$. CI measures the apparent bottleneck of the update loop, while CI^E discounts alignment that is purchased through intervention rather than through model adaptation. Generic changepoint detectors remain the stronger default alarms on passive streams. The diagnostic

role of CI^E is narrower and sharper: it targets a failure mode that passive alarms do not encode, namely the regime in which the world looks stable because the system is forcing it to look stable.

The scope is intentionally narrower than the opening contrast might suggest. The cube-root law is exact for the finite-memory averaging class analyzed here, not for every recursive or extrapolative architecture. The decomposition is motivated only for *action-effective* systems; passive observers are a genuine boundary case. Universal superiority over generic drift detectors on passive streams is neither claimed nor observed. The stability results remain conditional on explicit ISS hypotheses, with the Gaussian tracker serving as the constructive reference instance. The logged recommendation benchmark is interpreted at the proxy level: it is evidence for the usefulness of effort-aware diagnostics under logged exposure concentration, not a causal identification result about masking. Within that scope, temporal staleness imposes a real tracking floor, and under feedback, raw fit is not yet a safe diagnostic.

2 Coherence Diagnostics

The coherence-delay law constrains the representation term of the finite-memory averaging systems studied here, but closed-loop performance also depends on whether actions remain reproducible and whether adaptation remains effective. The ISS template of section 7 provides one way to score those three ingredients. This section packages them into a diagnostic built from the minimal decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$.

Definition 2.1 (Component Coherence Scores). Let $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ be an information system with ISS-Lyapunov functionals V_P , V_A , V_Φ as in section 7. Define the three *component coherence scores* at time t as:

$$\sigma_P(t) := \frac{1}{1 + V_P(t)} = \frac{1}{1 + \frac{1}{2}W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t))^2}, \quad (1)$$

$$\sigma_A(t) := \exp(-H(\mathcal{A}_t \mid \mathcal{P}(S, t))), \quad (2)$$

$$\sigma_\Phi(t) := \frac{1}{1 + V_\Phi(t)}. \quad (3)$$

Each score takes values in $(0, 1]$, equals 1 when the corresponding component is at its reference state, and approaches 0 as the component degrades.

Definition 2.2 (Coherence Index). The *Coherence Index* (CI) of a system $I(S, t)$ at time t , written formally as $\mathcal{C}(I(S, t), t)$ when the functional dependence must be shown explicitly, is

$$\mathcal{C}(I(S, t), t) := \min(\sigma_P(t), \sigma_A(t), \sigma_\Phi(t)) \in (0, 1].$$

The min aggregation is the correct choice given the Borromean property (theorem 7.2): system coherence is bounded by the weakest component, and strength in two components does not compensate for failure in the third.

Remark 2.3 (Bottleneck sensitivity and effort correction). The use of $\min(\sigma_P, \sigma_A, \sigma_\Phi)$ is bottleneck-sensitive: it diagnoses the weakest active subsystem, not overall system quality in the round. In action-effective environments, high actuation capacity can

keep σ_P artificially high by forcing the world toward a stale model. When such feedback masking is possible, a conservative extension is the *effort-corrected score*

$$\mathcal{C}^E(I(S, t), t) := \min(\sigma_P(t)e^{-\lambda E_t/E_0}, \sigma_A(t), \sigma_\Phi(t)),$$

where E_t is any monotone proxy for corrective actuation burden, $E_0 > 0$ is a reference scale, and $\lambda > 0$ controls sensitivity. It discounts apparent improvements in model-world alignment when those gains come from action-induced confounding rather than better state estimation. In the synthetic particle tracker, E_t is the realized coercive effort $\alpha|a_t - s_t^{\text{free}}|$; in deployed systems, action increments, energy expenditure, or control-surface usage can play the same role. For logged recommendation benchmarks, the preferred instantiation is the per-user KL divergence between the current exposure-tag distribution and the healthy random-exposure baseline distribution.

2.1 Estimable Score

Definition 2.1 requires $\mathcal{P}^*(S, t)$, the true oracle distribution of the environment, to compute $\sigma_P(t)$. In any deployed system, $\mathcal{P}^*$ is unobservable by construction: a system with complete knowledge of $\mathcal{P}^*$ would have nothing to estimate. We therefore introduce an estimable version of σ_P using an empirical approximation of $\mathcal{P}^*$.

Definition 2.4 (Empirical Potentiality Estimator). Let $\{s_{t-n+1}, \dots, s_t\}$ be the n most recent environmental signals observed by the system. The *empirical potentiality estimator* is the empirical measure:

$$\hat{\mathcal{P}}^*(S, t) := \frac{1}{n} \sum_{k=t-n+1}^t \delta_{s_k},$$

where δ_{s_k} is the Dirac measure at observation s_k .

Definition 2.5 (Estimable Score). The *debiased Sinkhorn divergence* between measures μ, ν is

$$S_\varepsilon(\mu, \nu) := T_\varepsilon(\mu, \nu) - \frac{1}{2}T_\varepsilon(\mu, \mu) - \frac{1}{2}T_\varepsilon(\nu, \nu),$$

where T_ε is the raw entropically regularized optimal transport cost. Unlike T_ε , the debiased divergence satisfies $S_\varepsilon(\mu, \mu) = 0$, restoring the metric property and enabling the clean exponent form of proposition 2.8, with ambient dimension, regularization, and support geometry absorbed into the constants rather than the displayed rate.

The *estimable component* $\hat{\sigma}_P$ replaces $\mathcal{P}^*$ with $\hat{\mathcal{P}}^*$ in eq. (1) using S_ε :

$$\hat{\sigma}_P(t) := \frac{1}{1 + \frac{1}{2}S_\varepsilon(\mathcal{P}(S, t), \hat{\mathcal{P}}^*(S, t))}.$$

The *estimable score* is

$$\hat{\mathcal{C}}(I(S, t), t) := \min(\hat{\sigma}_P(t), \sigma_A(t), \sigma_\Phi(t)).$$

Operationally, the three components sit on an observability ladder: ‘exact’ when the action channel or Lyapunov functional is constructively known, ‘identified’ when logging is rich enough for a direct estimator, and ‘proxy’ when the benchmark supports only an operational surrogate. In the present experiments, the Gaussian tracker is exact, the particle tracker is near the identified/surrogate boundary, and KuaiRand is proxy-level.

Remark 2.6 (Null and alternative calibration for Sinkhorn scores). Recent limit theorems for the empirical Sinkhorn divergence make an important calibration split explicit [Goldfeld et al., 2024]. Under the null $P = P^*$, the debiased statistic has an n^{-1} null fluctuation order, whereas under the drifted alternative regime in which the present estimation bound is used the statistical term follows the usual $n^{-1/2}$ scale. The practical implication is methodological rather than asymptotic: healthy reference windows and drifted evaluation windows should be calibrated separately rather than pooled into a single threshold rule. This is exactly the discipline used in the logged benchmark protocol, where thresholds are fit on healthy windows only and then evaluated on later regimes. The same line of work also supports bootstrap-based uncertainty quantification for Sinkhorn-type functionals, which matches the paired bootstrap intervals reported for KuaiRand.

Assumption 2.7 (W_2 -Lipschitz Drift). The oracle trajectory $t \mapsto \mathcal{P}^*(S, t)$ is W_2 -Lipschitz with drift rate $\zeta \geq 0$: for all $s, t \geq 0$,

$$W_2(\mathcal{P}^*(S, s), \mathcal{P}^*(S, t)) \leq \zeta |t - s|.$$

For the Gaussian family with common covariance, assumption 2.7 reduces to a Lipschitz bound on the mean trajectory.

Proposition 2.8 (Estimation Error Bound for $\hat{\sigma}_P$). *Let assumption 2.7 hold, and assume each oracle marginal $\mathcal{P}^*(S, t)$ is K -subgaussian. Let $\hat{\mathcal{P}}^*$ be the sliding-window empirical estimator with window n (definition 2.4), and let $\tau = (n - 1)/2$ be the effective lag of the centered window. Then under the entropic regularization of remark 5.3,*

$$\mathbb{E} |\sigma_P(t) - \hat{\sigma}_P(t)| \leq C_K n^{-1/2} + \zeta \tau,$$

where C_K absorbs the finite-sample Sinkhorn constant, the fixed regularization ε , the ambient geometry, and the uniform subgaussian envelope. This is a finite-memory upper bound for the sliding-window class, not a minimax statement over all adaptive estimators. In the experiments, the same constant is calibrated once per regime; its value changes with dimension and regularization, but the cube-root exponents do not.

Proof. Let

$$\bar{\mathcal{P}}^*(S, t) := \frac{1}{n} \sum_{k=t-n+1}^t \mathcal{P}^*(S, k)$$

be the oracle window-average law. The finite-sample term is the usual subgaussian Sinkhorn estimation error: by Genevay et al. [2019, Thm. 3], the debiased divergence between $\bar{\mathcal{P}}^*(S, t)$ and its empirical estimator is of order $n^{-1/2}$, with the regularization dependence absorbed into C_K .

The additional error comes from temporal staleness of the window. By convexity of W_2^2 in its second argument,

$$W_2(\mathcal{P}^*(S, t), \bar{\mathcal{P}}^*(S, t)) \leq \frac{1}{n} \sum_{k=t-n+1}^t W_2(\mathcal{P}^*(S, t), \mathcal{P}^*(S, k)).$$

Applying assumption 2.7,

$$\begin{aligned} W_2(\mathcal{P}^*(S, t), \bar{\mathcal{P}}^*(S, t)) &\leq \frac{\zeta}{n} \sum_{k=t-n+1}^t |t - k| \\ &= \zeta \frac{n-1}{2} = \zeta\tau. \end{aligned}$$

This is the explicit lag-bias term. Combining it with the $C_K n^{-1/2}$ statistical term yields the stated decomposition. Finally, the score map $x \mapsto 1/(1 + \frac{1}{2}x^2)$ is globally Lipschitz on $\mathbb{R}_{\geq 0}$, so the same order transfers to $\mathbb{E}|\sigma_P(t) - \hat{\sigma}_P(t)|$. $\square$

Remark 2.9 (Tightness of the lag-bias term). The lag term is not merely an artifact of upper-bounding. For monotone translation trajectories $\mathcal{P}^*(S, t) = \mathcal{P}_0(S - vt)$ with $\|v\| = \zeta$, the current law and the centered window-average law are separated by exactly the mean age of the window, so the temporal bias is of order $\zeta\tau$. The cube-root balance therefore reflects a real finite-memory limitation rather than a proof-only looseness.

Proposition 2.8 grounds $\hat{\mathcal{C}}$: the estimation error decreases with window size and increases with drift. This same $n^{-1/2}$ versus $\zeta\tau$ trade-off yields the tracking limit. When window lag is accounted for explicitly ($\tau = n/2$ for a centred window), the total estimation error decomposes as a sum of two opposing terms. Minimising over n yields the cube-root optimum for the averaging class, including the sliding-window and exponentially weighted schemes.

Definition 2.10 (Lag-Inclusive Estimation Error). For a sliding window of size n centred at lag $\tau = n/2$, the total expected estimation error is:

$$\mathcal{E}(n) = C_K n^{-1/2} + \frac{1}{2} \zeta n,$$

where the first term is the *variance* (decreasing in n) and the second is the *lag-bias* (increasing in n).

Proposition 2.11 (Optimal Window Under Drift). *Under definition 2.10, the estimation error $\mathcal{E}(n)$ is minimised at the optimal window*

$$n^* = \left(\frac{C_K}{\zeta} \right)^{2/3},$$

yielding the minimum achievable tracking error

$$\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}.$$

Proof. Setting $d\mathcal{E}/dn = 0$: $-\frac{1}{2}C_K n^{-3/2} + \frac{1}{2}\zeta = 0$, giving $n^* = (C_K/\zeta)^{2/3}$. Substituting back: $\mathcal{E}(n^*) = C_K(C_K/\zeta)^{-1/3} + \frac{1}{2}\zeta(C_K/\zeta)^{2/3} = C_K^{2/3}\zeta^{1/3} + \frac{1}{2}C_K^{2/3}\zeta^{1/3} = \frac{3}{2}C_K^{2/3}\zeta^{1/3}$. $\square$

Corollary 2.12 (EWMA Analogue). *Let the finite-memory estimator use exponential weights $w_j = \alpha(1-\alpha)^j$ for $j \geq 0$, so its effective lag is $\tau_\alpha = \sum_{j \geq 0} j w_j = (1-\alpha)/\alpha$ and its effective sample size is $n_{\text{eff}}(\alpha) = 1/\sum_{j \geq 0} w_j^2 = (2-\alpha)/\alpha$. Under assumption 2.7, the resulting finite-memory error obeys*

$$\mathcal{E}_{\text{EWMA}}(\alpha) \leq C_K \sqrt{\frac{\alpha}{2-\alpha}} + \zeta \frac{1-\alpha}{\alpha}.$$

Hence, in the small- α regime, $\alpha^* \propto (\zeta/C_K)^{2/3}$ and $\mathcal{E}_{\text{EWMA}, \min} \propto C_K^{2/3} \zeta^{1/3}$.

Proof. The variance term is the sliding-window rate with n replaced by the effective sample size $n_{\text{eff}}(\alpha)$. The lag term is the W_2 -Lipschitz drift rate multiplied by the effective age τ_α . For small α this becomes $\mathcal{E}_{\text{EWMA}}(\alpha) = O(C_K \alpha^{1/2} + \zeta \alpha^{-1})$. Balancing the two exponents gives $\alpha^* = O((\zeta/C_K)^{2/3})$, and substitution yields the same $\zeta^{1/3}$ minimum-error law as the sliding-window case. $\square$

Definition 2.13 (Causal Averaging Kernel). A *causal averaging kernel* is a sequence of weights $w = (w_0, w_1, \dots)$ such that $w_j \geq 0$ and $\sum_{j \geq 0} w_j = 1$. Its effective size and effective lag are

$$n_{\text{eff}}(w) := \frac{1}{\sum_{j \geq 0} w_j^2}, \quad \tau_{\text{eff}}(w) := \sum_{j \geq 0} j w_j.$$

Proposition 2.14 (Averaging-Kernel Lower Bound). *Let $X_t = \mu_t + \xi_t$ with $\xi_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d. and let the drift trajectory be the monotone translation model $\mu_t = \zeta t$ with $\zeta \geq 0$. For any causal averaging kernel w as in definition 2.13, the kernel estimator*

$$\hat{\mu}_t^{(w)} := \sum_{j \geq 0} w_j X_{t-j}$$

satisfies

$$\sqrt{\mathbb{E}(\hat{\mu}_t^{(w)} - \mu_t)^2} \geq c_1 \sigma n_{\text{eff}}(w)^{-1/2} \vee c_2 \zeta \tau_{\text{eff}}(w)$$

for universal constants $c_1, c_2 > 0$. Equivalently, the mean-squared error obeys the exact decomposition

$$\mathbb{E}(\hat{\mu}_t^{(w)} - \mu_t)^2 = \sigma^2 n_{\text{eff}}(w)^{-1} + \zeta^2 \tau_{\text{eff}}(w)^2,$$

and therefore the RMSE lower bound. In particular, optimizing over the averaging family yields the same cube-root exponent as proposition 2.11.

Proof. The estimator error decomposes as

$$\hat{\mu}_t^{(w)} - \mu_t = \underbrace{\zeta \sum_{j \geq 0} j w_j}_{\text{bias}} + \underbrace{\sum_{j \geq 0} w_j \xi_{t-j}}_{\text{noise}}.$$

Hence the bias magnitude is exactly $\zeta \tau_{\text{eff}}(w)$ and the noise term is Gaussian with variance $\sigma^2 \sum_{j \geq 0} w_j^2 = \sigma^2 / n_{\text{eff}}(w)$. Therefore

$$\mathbb{E}(\hat{\mu}_t^{(w)} - \mu_t)^2 = \zeta^2 \tau_{\text{eff}}(w)^2 + \sigma^2 n_{\text{eff}}(w)^{-1},$$

and taking square roots gives the stated lower bound, since $\sqrt{a+b} \geq a^{1/2} \vee b^{1/2}$ for nonnegative a, b . Optimizing the resulting trade-off over the averaging family recovers the same 2/3 window scaling and 1/3 error exponent as the upper bound. $\square$

The kernel result is exact but still tied to linear averaging. The minimax problem removes that linearity assumption on the estimator side by moving to all measurable rules that only observe a finite recent window. The price is that the signal class is now the full discrete ζ -Lipschitz family rather than the single monotone translation witness.

Proposition 2.15 (Window-Restricted Minimax Lower Bound). *Fix an integer $m \geq 1$ and suppose the estimator only observes the last m samples*

$$X_{t-m+1}, \dots, X_t, \quad X_s = \mu_s + \xi_s, \quad \xi_s \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2).$$

Let $\mathcal{L}_m(\zeta)$ be the discrete ζ -Lipschitz class

$$\mathcal{L}_m(\zeta) := \{(\mu_{t-m+1}, \dots, \mu_t) : |\mu_i - \mu_j| \leq \zeta|i - j| \text{ for all } i, j\}.$$

For any measurable estimator $\hat{\mu}_t = \delta_m(X_{t-m+1:t})$, define the worst-case root risk

$$R_m(\delta_m) := \sup_{\mu \in \mathcal{L}_m(\zeta)} \left(\mathbb{E}_\mu(\hat{\mu}_t - \mu_t)^2 \right)^{1/2}.$$

Then there exists a universal constant $c > 0$ such that

$$\inf_{\delta_m} R_m(\delta_m) \geq c \max \left\{ \sigma m^{-1/2}, \sup_{1 \leq h \leq m} \min(\sigma h^{-1/2}, \zeta h) \right\}.$$

In particular, if $1 \leq (\sigma/\zeta)^{2/3} \leq m$, then

$$\inf_{\delta_m} R_m(\delta_m) \geq c \sigma^{2/3} \zeta^{1/3}.$$

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$.

We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in $\mathcal{L}_m(\zeta)$. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\text{KL}(P_+, P_-) = \frac{1}{2\sigma^2} \sum_{j=1}^m (\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}} \right)^2 = \frac{1}{2}.$$

For any estimator δ_m , the plug-in test $\psi = \mathbf{1}\{\delta_m \geq 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam’s two-point lemma and Pinsker’s inequality, the average testing error is at least

$$\frac{1 - \text{TV}(P_+, P_-)}{2} \geq \frac{1 - \sqrt{\text{KL}(P_+, P_-)/2}}{2} \geq \frac{1}{4}.$$

Hence

$$R_m^* \geq \frac{\sigma}{8\sqrt{m}}$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \leq h \leq m$ and define

$$\beta_h := \min \left\{ \zeta, \frac{\sigma}{4h^{3/2}} \right\}, \quad \mu_{t-j}^{\pm, h} := \pm \beta_h (h - j)_+, \quad 0 \leq j \leq m - 1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to $\mathcal{L}_m(\zeta)$ because its slope magnitude is at most $\beta_h \leq \zeta$. The endpoint gap is

$$|\mu_t^{+, h} - \mu_t^{-, h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\text{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2} \sum_{j=0}^{m-1} (2\beta_h(h-j)_+)^2 \leq \frac{2\beta_h^2}{\sigma^2} \sum_{r=1}^h r^2 \leq \frac{2\beta_h^2}{\sigma^2} h^3 \leq \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\beta_h h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1 - \text{TV}(P_{+,h}, P_{-,h})}{2} \geq \frac{1 - \sqrt{\text{KL}(P_{+,h}, P_{-,h})/2}}{2} \geq \frac{3}{8}.$$

Therefore

$$R_m^* \geq \frac{3}{8} \beta_h h \geq \frac{3}{32} \min\{\zeta h, \sigma h^{-1/2}\}$$

which is again a fully explicit numerical lower bound. Since h was arbitrary,

$$R_m^* \geq \frac{3}{32} \sup_{1 \leq h \leq m} \min\{\zeta h, \sigma h^{-1/2}\}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \leq h_* := (\sigma/\zeta)^{2/3} \leq m$. Choose an integer $\bar{h} \in [h_*/2, h_*]$. Then

$$\zeta \bar{h} \geq \frac{1}{2} \zeta h_* = \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and

$$\sigma \bar{h}^{-1/2} \geq \sigma h_*^{-1/2} = \sigma^{2/3} \zeta^{1/3}.$$

Therefore

$$\min\{\sigma \bar{h}^{-1/2}, \zeta \bar{h}\} \geq \frac{1}{2} \sigma^{2/3} \zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \geq \frac{3}{64} \sigma^{2/3} \zeta^{1/3}$$

so one admissible explicit constant in the critical-window regime is $c = 3/64$. This proves the claim. $\square$

In other words, the displayed cube-root floor is operational exactly in the critical-window regime $1 \leq (\sigma/\zeta)^{2/3} \leq m$: outside that range, the problem is either noise-dominated or too window-limited for the minimax obstruction to reach the balancing scale.

Remark 2.16 (Interpretation: a finite-memory floor). Proposition 2.11 says that continuous drift imposes a nonzero floor for the finite-memory averaging class: $\mathcal{E}_{\min} \propto \zeta^{1/3}$ and $n^* \propto \zeta^{-2/3}$. In practice this becomes a self-tuning rule, $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$, with the critical-window lower bound in proposition 2.15 showing that the same obstruction persists for any estimator restricted to the last m samples.

3 Coercive Masking and Effort-Aware Correction

Closed-loop systems admit a specific failure mode beyond ordinary drift. A controller can keep the realized trajectory simple either because it tracks the world well or because it pushes the world toward a stale model. The latter mechanism is invisible to a diagnostic that looks only at model-world fit. This section defines that masking regime and the effort-aware correction used in the active and logged benchmarks.

Definition 3.1 (Coercive Regime). An information system $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ operates in the *coercive regime* if:

- (i) $\mathcal{A}_t$ is action-effective (definition 6.2) with coupling coefficient $\alpha > 0$: the transition kernel is $P(s_{t+1} | s_t, o) = \mathcal{N}(\mu^*(t) + \alpha(o - \mu^*(t)), \sigma_w^2)$;
- (ii) $\mathcal{P}$ is static (proposition 8.1): $\hat{\mu}_t \equiv \hat{\mu}_0$ and $\mathcal{A}_t(\mathcal{P}) \equiv \hat{\mu}_0$;
- (iii) the exogenous drift is bounded: $|\zeta| < \alpha \cdot |\hat{\mu}_0 - \mu_0|$.

In the coercive regime, the system’s stale actions continuously pull μ^* toward the frozen estimate $\hat{\mu}_0$: $\mu^*(t+1) = \mu^*(t) + \zeta + \alpha(\hat{\mu}_0 - \mu^*(t))$, which is a mean-reverting process with equilibrium $\hat{\mu}_0 + \zeta/\alpha$. When $|\zeta/\alpha|$ is small, μ^* is held near $\hat{\mu}_0$, suppressing V_P and making the system appear healthy under the standard score.

Corollary 3.2 (Coercive Masking). *In the coercive regime (definition 3.1), the standard score $\sigma_P(t) = 1/(1 + V_P(t))$ can fail to detect FM-1: $\sigma_P(t) \approx 1$ for all $t \leq t_{\text{mask}}$, where $t_{\text{mask}} = O(\alpha^{-1} \log(\alpha/\zeta))$ is the coercive horizon. Beyond t_{mask} , exogenous drift overwhelms the coercive capacity and V_P diverges.*

Proof sketch. Under condition (iii) of definition 3.1, the gap $\delta_t = \mu^*(t) - \hat{\mu}_0$ satisfies $\delta_{t+1} = (1 - \alpha)\delta_t + \zeta$. The fixed point is $\delta^* = \zeta/\alpha$, so $V_P(t) \leq \frac{1}{2}(\zeta/\alpha)^2 + \sigma_w^2/(2\alpha)$ in expectation, masking FM-1 while the coercive condition holds. Once the drift accumulates beyond the available control authority, the frozen model can no longer hold the environment in place and the underlying mismatch reappears.

Remark 3.3 (Compression by adaptation vs. compression by forcing). From the algorithmic-regulation viewpoint, coercive masking matters because it can improve realized compressibility without improving the model. The external trajectory becomes simpler, but for the wrong reason: the controller is forcing the world into alignment rather than updating its internal state to match a changing world.

Definition 3.4 (Effort-Adjusted Coherence Index). The *effort signal* at time t is any monotone proxy for the corrective actuation burden required to keep the world aligned with the system’s present output. A generic observable surrogate is

$$E_t^{\text{obs}} := |\mathcal{A}_t(\mathcal{P}) - \mathcal{A}_{t-1}(\mathcal{P})|,$$

the magnitude of change in the system’s actuation. In controlled synthetic settings, a sharper proxy is the realized coercive effort

$$E_t^{\text{sim}} := \alpha |a_t - s_t^{\text{free}}|,$$

where s_t^{free} is the state that would have arisen before applying control.

The *Effort-Adjusted Coherence Index* (CI^E), written formally as $\mathcal{C}^E(I(S, t), t)$ when the functional dependence must be shown explicitly, is

$$\mathcal{C}^E(I(S, t), t) := \min(\sigma_P(t) \cdot e^{-\lambda E_t/E_0}, \sigma_A(t), \sigma_\Phi(t)),$$

where $E_0 > 0$ is a reference effort scale and $\lambda > 0$ is a sensitivity parameter. CI^E penalizes increasing actuation effort, correcting for action-induced confounding: it discounts apparent coherence when the environment is being held in place rather than genuinely estimated. In action-effective settings where coercive masking is plausible, CI^E is the primary closed-loop diagnostic and CI is the corresponding apparent-fit baseline.

Definition 3.5 (Masking Gap). The *masking gap* at time t is

$$M_t := \mathcal{C}(I(S, t), t) - \mathcal{C}^E(I(S, t), t) \geq 0.$$

It quantifies how much apparent coherence is lost once corrective effort is charged explicitly. Large M_t indicates that the observed fit is being maintained by intervention rather than by an up-to-date model.

Remark 3.6 (Calibration guidance for E_t , E_0 , and λ). The effort proxy should follow the strongest observable hierarchy available: E_t^{sim} in controlled simulators, E_t^{obs} from direct actuator burden in deployed systems, and E_t^{log} from logged exposure or policy divergence when only observational traces are available.

The reference scale E_0 should be set to the typical healthy effort level of the domain, not to a global constant. In practice this means using a nominal controller effort in simulation or the median healthy effort over a temporally held-out prefix in logged data. In the particle tracker we use the prior-scale effort floor, in KuaiRand we use the median healthy KL divergence from the random-exposure baseline, and in deployed control systems the right default is the nominal actuator burden under steady operation.

The penalty λ controls how aggressively apparent coherence is discounted. Across the particle masking grid and the KuaiRand sensitivity sweep, values in the range $\lambda \in [2, 3]$ provide the most stable trade-off, with $\lambda = 3$ serving as the default: $\lambda < 1$ under-penalizes masking, while $\lambda = 4$ tends to be more aggressive than the added separation usually justifies.

These sweeps function as the current ablations for proxy strength and effort sensitivity: they show that the main conclusions are stable across a useful operating range, while still making clear that calibration is part of the diagnostic rather than a nuisance parameter hidden from the analysis.

The empirical evidence for coercive masking comes from the particle-tracker experiment (section 4.5). In the passive FM-1 regime, CI and CI^E coincide because no realized forcing of the world occurs, so $M_t = 0$. Under moderate coupling ($\alpha = 0.3$, $\lambda = 3.0$), the same frozen model reaches $CI = 0.949 \pm 0.012$ while $CI^E = 0.697 \pm 0.046$, so the mean masking gap is $M_t \approx 0.252$, showing that the apparent coherence is being purchased by intervention rather than adaptation.

4 Empirical Validation

A Gaussian tracker provides an analytically tractable closed-loop tracking model that instantiates the three roles $(\mathcal{P}, \mathcal{A}, \Phi)$ and admits closed-form computation of all Lyapunov functionals.

The active and logged results come first, because they are where effort-aware diagnostics matter most. The supporting studies then recover the cube-root scaling, ablate the three-role diagnostic, and place the method against passive drift alarms. Code and reproduction instructions are provided in appendix A.

4.1 Logged and Active Masking

Two results carry the empirical argument. On KuaiRand, healthy-only calibration raises bubble detection from 0.458 with CI to 0.692 with CI^E , with paired bootstrap intervals excluding zero for the user-level gain; all generic detectors in that comparison consume the same effort-adjusted scalar signal, so the comparison is about detector behavior rather than detector-specific inputs. On the active particle benchmark, CI^E detects masking onset in all 12 runs with median delay 5.0, whereas ADWIN detects 7 of 12 runs with median delay 279.0.

These are the regimes in which the diagnostic claim matters: low observed error can be maintained by intervention, and passive alarms do not directly price that intervention burden. The detailed logged and active protocols appear in sections 4.5 and 4.8. What follows tests the finite-memory law, the score construction, and the passive-regime split.

4.2 Formalization

The Gaussian tracker extends example 7.4. Let the environment produce observations from a drifting Gaussian:

$$s_t \sim \mathcal{N}(\mu^*(t), \sigma_{\text{obs}}^2), \quad \mu^*(t) = \mu_0 + \zeta t + \sigma_w B_t,$$

where B_t is standard Brownian motion. The tracker components are:

- $\mathcal{P}(S, t) = \mathcal{N}(\hat{\mu}_t, \hat{\sigma}_t^2)$: Gaussian belief state.
- $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$: deterministic mean extraction ($H(\mathcal{A}_t | \mathcal{P}) = 0$, $\delta = 0$).
- Φ_t : Kalman measurement update (see example 7.4).

All three Lyapunov functionals are available in closed form (example 7.4), which makes the tracker a convenient test bed for the trade-off and diagnostic claims.

Lemma 4.1 (The Gaussian Tracker Satisfies the ISS Hypotheses). *The Gaussian tracker with $\sigma_w > 0$ and $\sigma_{\text{obs}} > 0$ satisfies all four ISS hypotheses (H-ISS-P/A/ Φ /Env) of theorem 7.2:*

- (i) **H-ISS-P**: $V_P(t) = \frac{1}{2}W_2(\mathcal{P}, \mathcal{P}^*)^2 = \frac{1}{2}(\hat{\mu}_t - \mu^*(t))^2$ is a valid ISS-Lyapunov functional; the Kalman update contracts V_P at rate $1 - K_t$ per step, where $K_t \in (0, 1)$ is the Kalman gain.
- (ii) **H-ISS-A**: $V_A = H(\mathcal{A}_t | \mathcal{P}) = 0$ since $\mathcal{A}_t = \hat{\mu}_t$ is deterministic; H-ISS-A is satisfied trivially.
- (iii) **H-ISS- Φ** : the Kalman recursion for $\hat{\sigma}_t^2$ converges to the steady-state variance $\hat{\sigma}_\infty^2 = \frac{1}{2}(\sigma_w^2 + \sqrt{\sigma_w^4 + 4\sigma_w^2\sigma_{\text{obs}}^2})$, a stable fixed point; $V_\Phi = |\hat{\sigma}_t^2 - \hat{\sigma}_\infty^2|$ decays geometrically with rate $1 - K_\infty$.
- (iv) **H-ISS-Env**: the environmental drift rate ζ is bounded by assumption; H-ISS-Env holds whenever $\zeta < (1 - K_\infty)\sigma_{\text{obs}}$.

Proof. Items (i)–(iii) follow directly from the Kalman filter’s convergence properties [Sontag, 1989, Mironchenko and Prieur, 2020]. Item (iv) is the H-ISS-Env condition of theorem 7.2 applied to the scalar Gaussian case; the contraction rate equals the steady-state Kalman gain $K_\infty = \hat{\sigma}_\infty^2 / (\hat{\sigma}_\infty^2 + \sigma_{\text{obs}}^2)$. $\square$

All experiments use `uv sync` from the repository for exact environment reproduction. The controlled Gaussian studies use the parameter values stated in their protocol paragraphs, while the ELEC2 benchmark keeps the same fixed warning level and Page-Hinkley calibration throughout.

4.3 Experiment A: Component Ablation

Protocol. We run $T = 500$ steps with $\zeta = 0.02$, $\sigma_w = 0.1$, $\sigma_{\text{obs}} = 1.0$, $\text{seed} = 42$, under four conditions:

C1. Full model: standard Gaussian tracker.

C2. FM-1 (static $\mathcal{P}$): $\hat{\mu}_t \equiv 0$ for all t ; Kalman update disabled.

C3. FM-2 (non-det. $\mathcal{A}$): $\mathcal{A}_t = \hat{\mu}_t + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, 0.25)$.

C4. FM-3 (absent Φ): Kalman update disabled; no feedback.

Environment: pure drift with no action feedback ($\alpha = 0$), so the world drifts independently of the system’s actuation. The effect of action-environment coupling is addressed in the Coercive Masking Corollary (corollary 3.2).

Results. Table 1 summarizes the tail-mean Lyapunov functionals and component coherence scores.

Table 1: Ablation results ($T = 500$, $\zeta = 0.02$, influence= 0, 42 seeds). Cells report mean $\pm$ standard deviation over the final 100 steps.

Condition	$\bar{V}_P$	$\bar{V}_A$	$\bar{V}_\Phi$	$\bar{\sigma}_P$	$\bar{\sigma}_A$	$\bar{\sigma}_\Phi$
Full model	0.051 $\pm$ 0.020	0.000 $\pm$ 0.000	0.010 $\pm$ 0.000	0.955 $\pm$ 0.017	1.000 $\pm$ 0.000	0.990 $\pm$ 0.000
FM-1 (static $\mathcal{P}$)	41.156 $\pm$ 16.005	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.034 $\pm$ 0.046	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
FM-2 (noisy $\mathcal{A}$)	0.051 $\pm$ 0.020	0.974 $\pm$ 0.139	0.010 $\pm$ 0.000	0.955 $\pm$ 0.017	0.579 $\pm$ 0.031	0.990 $\pm$ 0.000
FM-3 (absent Φ)	1.972 $\pm$ 3.680	0.000 $\pm$ 0.000	5.400 $\pm$ 0.000	0.573 $\pm$ 0.259	1.000 $\pm$ 0.000	0.157 $\pm$ 0.000

The results confirm the stability template of theorem 7.2: V_{total} is lowest under the full model. FM-1 and FM-3 both fail, but via different components: the score correctly identifies $\mathcal{P}$ as the bottleneck in FM-1 ($\bar{\sigma}_P = 0.040$) and Φ as the bottleneck in FM-3 ($\bar{\sigma}_\Phi = 0.157$), while FM-2 is selectively degraded through $\mathcal{A}$ ($\bar{\sigma}_A = 0.530$). The passive full-model run remains sharply concentrated near the equilibrium ($\bar{V}_{\text{total}} = 0.027$), whereas FM-1 rises to $\bar{V}_{\text{total}} = 24.259$, providing the diagnostic precision claimed in definition 2.2.

V_{total} is not sample-path monotone, consistent with theorem 7.2 which guarantees non-increasing *expectation* under the ISS hypotheses, not sample-path monotonicity of the stochastic process.

4.4 Experiment B: Sinkhorn Sample Complexity and the Coherence-Delay U-Curve

Protocol. We use the Gaussian tracker at stationary regime ($T = 5000$, 50 seeds, $\sigma_{\text{obs}} = 1$) and compute $\hat{\sigma}_P(t)$ using sliding-window estimators with $n \in$

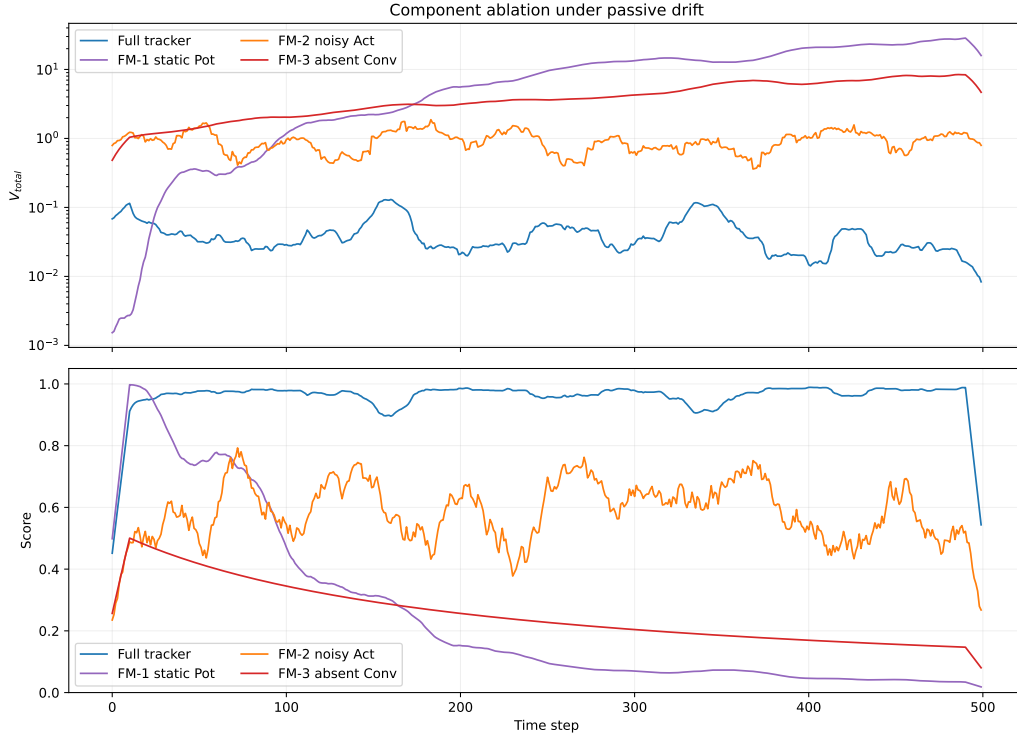


Figure 1: Component ablation: V_{total} (log scale, 20-step rolling mean, top) and the Coherence Index (CI, bottom) under the four conditions. The full model is the only condition where CI remains bounded away from 0. FM-1 and FM-3 both diverge in V_{total} but are distinguished by different bottleneck components: $\sigma_P \rightarrow 0$ for FM-1 (stale $\mathcal{P}$) and $\sigma_\Phi \rightarrow 0$ for FM-3 (absent Φ).

$\{5, 10, 25, 50, 100\}$. Mean absolute error (MAE) $|\sigma_P - \hat{\sigma}_P|$ is measured in the tail (last 40% of T).

The lag-drift tradeoff and the Coherence-Delay Principle. The estimation error has two competing terms (definition 2.10):

$$\mathcal{E}(n) \leq \underbrace{C_K n^{-1/2}}_{\text{variance term}} + \underbrace{\zeta \cdot \frac{n}{2}}_{\text{lag-bias term}}. \quad (4)$$

At large n or fast drift, the lag-bias term dominates and the total error increases with n , hiding the $O(n^{-1/2})$ variance rate.

To validate proposition 2.11 empirically, we additionally run a U-curve sweep: for each of four drift rates $\zeta \in \{0.001, 0.005, 0.01, 0.05\}$, we measure the raw tracking error $|\mu^*(t) - \bar{s}_{t-n:t}|$ across $n \in \{5, 10, 20, 50, 75, 100, 150, 200, 300, 500\}$ in a passive environment (influence = 0) over $T = 6,000$ steps and 20 seeds. For each drift level we extract the empirical minimum over the window grid and then fit the log-log slope of $\mathcal{E}_{\min}$ against ζ by ordinary least squares.

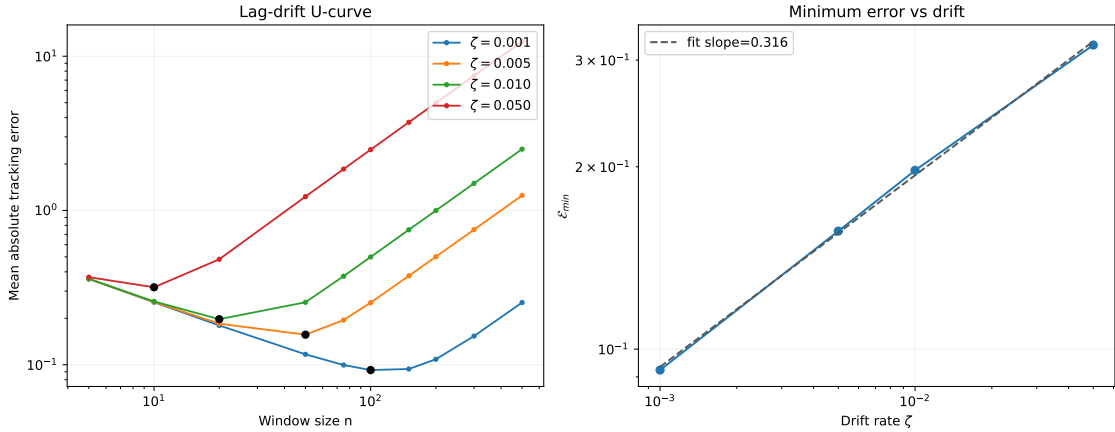


Figure 2: Coherence-delay U-curve validation (proposition 2.11). *Left*: empirical tracking error vs. window size n for four drift rates (log scale; dotted lines are scaled theoretical curves; stars mark empirical n^*). Each curve shows the predicted U-shape: variance-dominated decay for small n , lag-dominated growth for large n . *Right*: $\mathcal{E}_{\min}$ vs. ζ on log-log scale. Fitted slope = 0.316 (95% CI [0.287, 0.345]); theoretical prediction = 1/3.

Results. Figure 2 confirms the two predictions of proposition 2.11. The U-curves show the characteristic variance-lag tradeoff for each ζ , with empirical n^* shrinking as drift increases (from $n^* \approx 100$ at $\zeta = 0.001$ to $n^* \approx 10$ at $\zeta = 0.05$). On the same four-point grid, a log-log fit of n^* against ζ gives slope -0.611 , already close to the predicted $-2/3$ memory law despite the coarse window discretization. The log-log slope of $\mathcal{E}_{\min}$ vs. ζ is **0.316** (95% CI [0.287, 0.345]), close to the theoretical $1/3$ scaling for this finite-window estimator class. Table 2 reports the extracted minima.

The ratio $\mathcal{E}_{\min}/\zeta^{1/3}$ remains close to constant (≈ 0.91 across the four drift levels), confirming the expected $\zeta^{1/3}$ scaling despite finite-window effects.

Table 2: Empirical validation of the Coherence-Delay bound ($T = 6,000$, 20 seeds, passive environment).

ζ	n_{obs}^*	$\mathcal{E}_{\min}$	$\mathcal{E}_{\min}/\zeta^{1/3}$
0.001	100	0.0924	0.9236
0.005	50	0.1566	0.9159
0.010	20	0.1973	0.9158
0.050	10	0.3177	0.8623

EWMA baseline and finite-memory universality. To test whether the observed cube-root law is specific to the uniform sliding-window estimator, we ran the same Gaussian drift sweep with an exponentially weighted moving average (EWMA) $\hat{\mu}_t = \alpha s_t + (1 - \alpha)\hat{\mu}_{t-1}$ over $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5\}$.

Table 3: EWMA baseline on the Gaussian tracker: optimal forgetting factor by drift rate.

ζ	α^*
0.001	0.02
0.005	0.05
0.010	0.10
0.050	0.20

The minimum-error EWMA slope is **0.3173** on a log-log fit of $\mathcal{E}_{\min}$ vs. ζ , essentially identical to the sliding-window slope **0.316** in fig. 2. The optimal forgetting factor follows log-log slope **0.6011**, close to the $2/3$ prediction from corollary 2.12. Empirically, the optimal forgetting factors increase with drift exactly as finite effective memory theory predicts, supporting the claim that the cube-root law is a property of the finite-memory class rather than an artifact of uniform windows. The EWMA slope is extracted by the same log-log fitting protocol as the sliding-window case, now using the minimum error over the tested α grid at each drift level.

This is also the right place to separate the present theorem from stronger parametric filters. Forgetting-factor RLS and Kalman predictors address a closely related responsiveness-versus-variance trade-off, but they belong to a different estimator class: they exploit explicit state-space or parametric structure rather than only finite-memory averaging. They are appropriate empirical baselines in the passive benchmarks, but not counterexamples to the cube-root law proved here. In masking regimes that rely on suppressing innovations, a model-based filter can look artificially healthy when the controller keeps its innovation signal small, whereas CI^E charges the effort needed to maintain that apparent fit.

In a correctly specified scalar parametric tracker with forgetting factor λ , the same engineering intuition can be written heuristically as a variance-lag balance,

$$\mathcal{E}_{\text{RLS}}^2(\lambda) \asymp \frac{\sigma^2(1 - \lambda)}{1 + \lambda} + \frac{\zeta^2}{(1 - \lambda)^2},$$

so the optimal responsiveness satisfies $1 - \lambda^* \asymp (\zeta/\sigma)^{2/3}$ and the corresponding root-risk scale is again of order $\sigma^{2/3}\zeta^{1/3}$; see Ljung [1999] for the classical forgetting-factor

perspective. We do not prove that formula here, and we do not claim that all such parametric filters satisfy the present theorem. The point is narrower: even when a parametric filter shares the same exponent under correct specification, that does not collapse the distinction in estimator class or remove the masking failure mode that motivates CI^E .

For the score estimation rate, we use slow drift ($\zeta = 0.001$, $\sigma_w = 0.035$) to stay near the variance-dominated regime while retaining a visible lag term at $n = 100$. The observed log-log slope is -0.436 (95% CI $[-0.865, -0.008]$; target -0.5); the deviation reflects the onset of lag-bias at the largest window and residual autocorrelation induced by the slow-drift regime (table 4 and fig. 3).

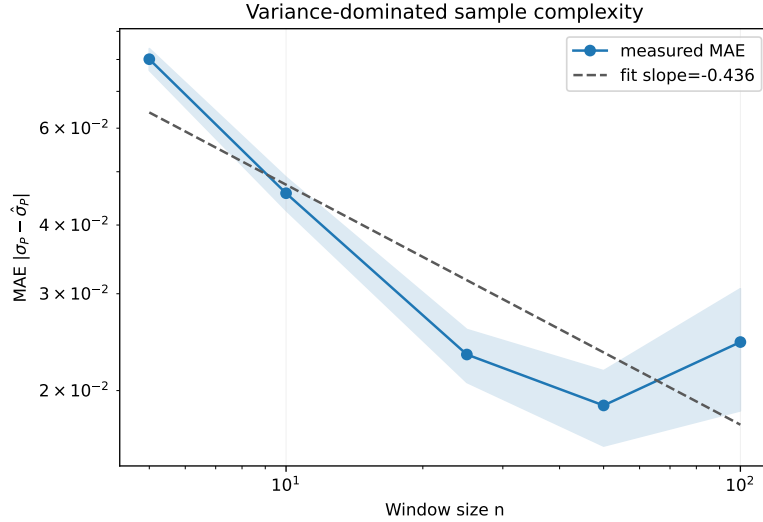


Figure 3: Sinkhorn sample complexity in the variance-dominated regime ($\zeta = 0.001$, $n \leq 100$, 50 seeds, $T = 5000$). Empirical log-log slope = -0.436 (95% CI $[-0.865, -0.008]$); theoretical target -0.5 . This is the variance-only regime of the U-curve in fig. 2.

Table 4: Sinkhorn estimation error vs window size ($\zeta = 0.001$, 50 seeds, $T = 5000$; mean $\pm$ standard deviation).

n	MAE $ \sigma_P - \hat{\sigma}_P $	std
5	0.0801	0.0036
10	0.0457	0.0032
25	0.0233	0.0026
50	0.0188	0.0029
100	0.0245	0.0061
Log-log slope: -0.436 (target: -0.5)		

Computational budget. While evaluating exact Wasserstein distances scales poorly, the debiased Sinkhorn divergences used for $\hat{\sigma}_P$ and $\hat{\sigma}_A$ are highly parallelizable. Figure 4 and table 5 profile the computational cost across sequence lengths

$n \in \{25, 50, 100\}$, ambient dimensions $d \in \{2, 8, 32, 64, 128, 256\}$, and regularization parameters $\varepsilon \in \{0.02, 0.05, 0.2, 1.0\}$. The metrics isolate the pairwise distance matrix computation and Sinkhorn iteration loop. On this grid the solver remains stable as dimension increases, but fixed ε is not portable: larger regularization is consistently cheaper and also gives lower absolute bias on the Gaussian clouds we tested. For instance, at $d = 8$ and $n = 100$, mean solve time falls from 7.341 ms at $\varepsilon = 0.05$ to 0.661 ms at $\varepsilon = 1.0$, while mean absolute bias falls from 0.6102 to 0.1383; at $d = 256$ and $n = 100$, the corresponding pair is 0.799 ms / 1.7778 at $\varepsilon = 0.05$ versus 0.571 ms / 0.1446 at $\varepsilon = 1.0$. The practical message is that intrinsic-dimension-aware calibration matters even when the same S_ε machinery is used throughout. These are operational drift-comparison sweeps, not null-calibration experiments; the empirical Sinkhorn limit theory indicates that null fluctuations obey a different order and should be handled separately when setting thresholds [Goldfeld et al., 2024].

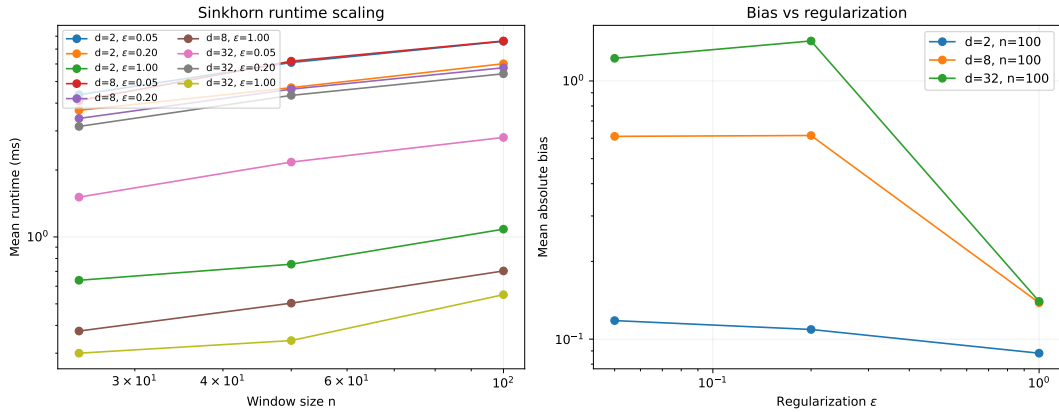


Figure 4: Sinkhorn runtime and bias trade-off on Gaussian clouds. *Left*: mean runtime versus window size for each combination of dimension and regularization. *Right*: mean absolute bias versus regularization at the largest window size ($n = 100$); the sweep extends to $d = 256$ and $\varepsilon = 0.02$.

Table 5: Representative Sinkhorn runtime points (Gaussian clouds). The grid now extends to $d = 256$ and $\varepsilon = 0.02$.

Setting	Mean runtime (ms)	Mean abs. bias	Mean iters	Pairwise/s
$d = 8, n = 25, \varepsilon = 0.02$	1.436	0.9522	250.0	435,296
$d = 8, n = 100, \varepsilon = 0.05$	7.341	0.6102	250.0	1,362,217
$d = 8, n = 100, \varepsilon = 1.0$	0.661	0.1383	14.5	15,121,765
$d = 128, n = 100, \varepsilon = 0.05$	1.179	1.6354	81.5	8,481,666
$d = 256, n = 100, \varepsilon = 0.05$	0.799	1.7778	35.5	12,511,617
$d = 256, n = 100, \varepsilon = 1.0$	0.571	0.1446	6.2	17,519,521

4.5 Experiment F: Feedback Masking in the Particle Tracker

This experiment instantiates corollary 3.2 and definition 3.4 in the non-Gaussian, non-linear particle tracker. The environment evolves according to the scalar particle-filter simulator implemented in `experiments/particle/model.py`, but now with an explicit action-coupling parameter $\alpha \in [0, 1]$ so that the actuation can pull the latent

state toward the stale controller output. In the synthetic setting we operationalise the effort proxy as the *realized coercive effort*

$$E_t^{\text{sim}} = \alpha |a_t - s_t^{\text{free}}|,$$

where s_t^{free} is the latent state that would have arisen at time t before applying control. This distinguishes mere action-world mismatch from actual world-forcing.

Protocol. We freeze $\mathcal{P}$ (FM-1) and compare two regimes: passive ($\alpha = 0$) and coercive ($\alpha > 0$). The main single-run diagnostic uses $\alpha = 0.3$ and effort penalty $\lambda = 3.0$. We then sweep $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$ and $\lambda \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ over 8 seeds, exporting all raw and aggregated metrics from the experiment harness. To keep the main text readable, table 6 reports a five-regime masking matrix at the default $\lambda = 3.0$: passive, weak, mild, moderate, and strong coercive coupling.

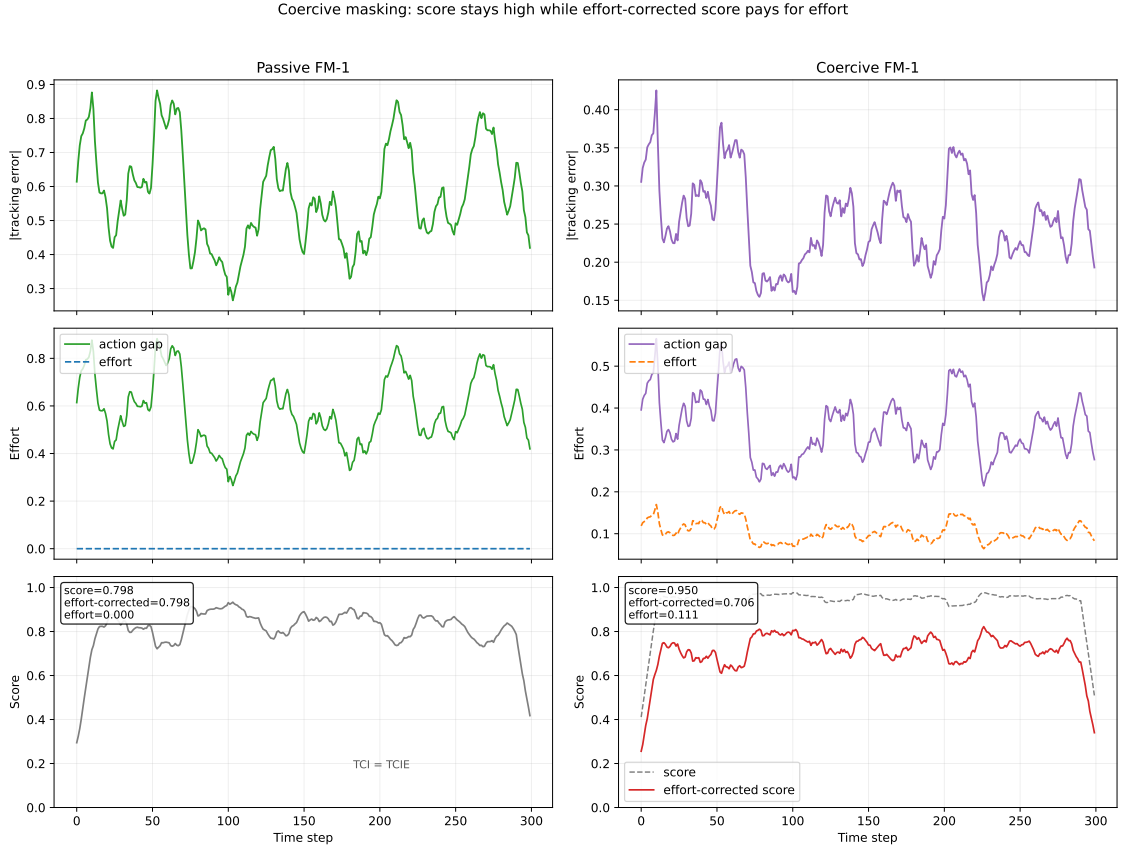


Figure 5: Synthetic feedback masking in the particle tracker. Left column: passive FM-1, where effort remains at zero and CI and CI^E coincide. Right column: coercive FM-1, where the action gap remains nonzero, realized coercive effort appears once influence is positive, and CI^E drops once effort is penalized.

Results. The matrix makes the masking mechanism explicit. As coupling increases, the stale FM-1 controller looks progressively healthier under the plain score: CI rises from 0.795 in the passive regime to 0.999 under strong coercion. The effort-aware diagnostic moves in the opposite direction: CI^E falls from 0.795 to 0.518, and

Table 6: Five-regime masking matrix for frozen- $\mathcal{P}$ FM-1 in the particle tracker at $\lambda = 3.0$ (8 seeds). Cells report tail means.

Regime	α	effort	CI	CI^E	M_t
Passive	0.0	0.000	0.795	0.795	0.000
Weak coercion	0.1	0.054	0.872	0.757	0.115
Mild coercion	0.2	0.090	0.920	0.726	0.194
Moderate coercion	0.3	0.118	0.949	0.697	0.252
Strong coercion	0.9	0.266	0.999	0.518	0.481

the masking gap grows from 0 to 0.481. This is the closed-loop signature we care about: apparent coherence is improving while effort-aware coherence is deteriorating.

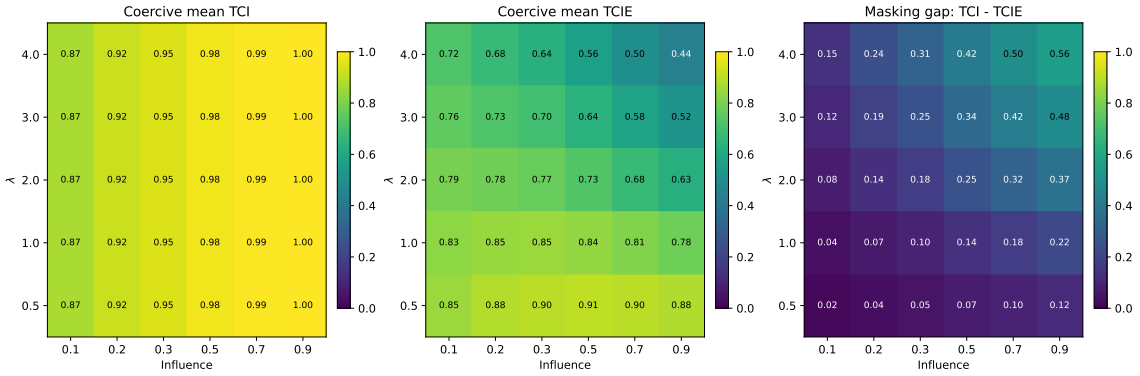


Figure 6: Multi-seed parameter sweep for feedback masking in the particle tracker (8 seeds). Left: mean CI. Middle: mean CI^E . Right: masking gap $M_t = CI - CI^E$. The standard index monotonically rises with influence, while the masking gap widens with both influence and effort penalty λ .

The sweep confirms that the correction is not an artifact of one parameter choice. At fixed $\lambda = 3.0$, the masking gap increases from 0.115 at $\alpha = 0.1$ to 0.481 at $\alpha = 0.9$, with a clear intermediate effect already at $\alpha = 0.2$ (gap = 0.194). At fixed $\alpha = 0.3$, increasing λ from 0.5 to 4.0 expands the gap from 0.051 to 0.313. Thus, stronger coercive capacity and stronger effort penalisation separate apparent from genuine coherence in the expected direction. The particle tracker closes the empirical loop on corollary 3.2: the standard score can be masked by control, whereas CI^E remains sensitive to the intervention cost that creates the mask. Combined with the calibration grid in fig. 8, this supports the practical rule: set E_0 to the healthy effort scale and choose λ in the range 2–3, with $\lambda = 3$ as a strong default.

To test whether this advantage survives comparison with a generic change detector, we build an active benchmark with two known regime transitions: healthy tracking for $t < 200$, feedback masking for $200 \leq t < 400$, and coercive collapse for $t \geq 400$. The masking phase freezes $\mathcal{P}$ while maintaining moderate coupling ($\alpha = 0.3$); the collapse phase sharply increases exogenous drift. We compare thresholded CI, thresholded CI^E , ADWIN, PageHinkley, KSWIN, and NoDrift on the raw observation stream, giving the generic detectors a fully observable signal rather than a model-derived residual.

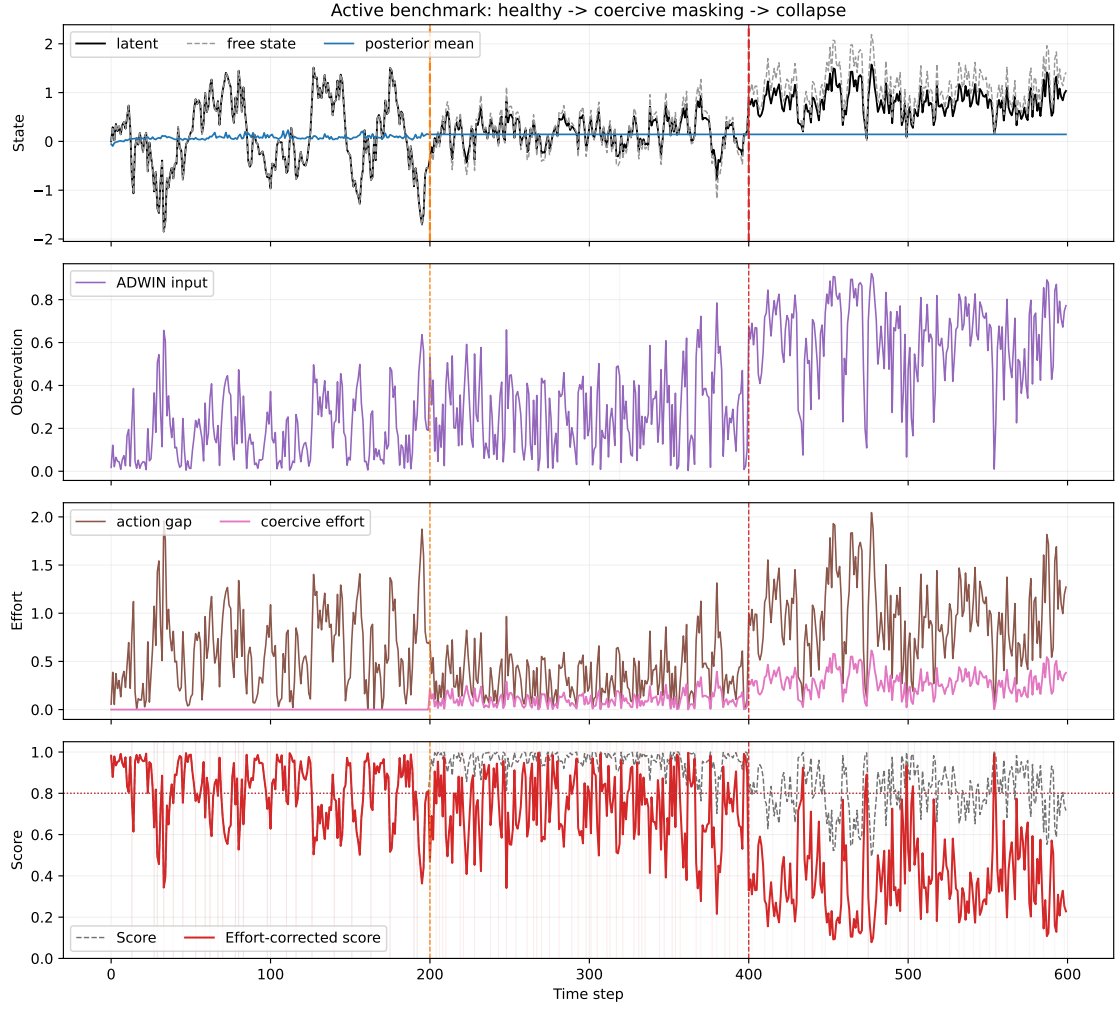


Figure 7: Active benchmark in the particle tracker with two known phase changes. Orange line: masking onset at $t = 200$. Red line: collapse onset at $t = 400$. ADWIN receives the raw observation stream. CI and CI^E are thresholded at 0.8.

Table 7: Active benchmark summary in the particle tracker (12 seeds). Detection is measured against the known onset of feedback masking and the known onset of coercive collapse.

Phase	Detector	Detections	Rate	Median delay
Masking onset	CI	12/12	1.00	25.5
Masking onset	CI ^E	12/12	1.00	5.0
Masking onset	ADWIN	7/12	0.58	279.0
Masking onset	PageHinkley	12/12	1.00	68.5
Masking onset	KSWIN	1/12	0.08	118.0
Masking onset	NoDrift	0/12	0.00	NA
Collapse onset	CI	12/12	1.00	2.0
Collapse onset	CI ^E	12/12	1.00	0.0
Collapse onset	ADWIN	5/12	0.42	111.0
Collapse onset	PageHinkley	12/12	1.00	37.5
Collapse onset	KSWIN	0/12	0.00	NA
Collapse onset	NoDrift	0/12	0.00	NA

The benchmark isolates the point that passive ELEC2 cannot address. ADWIN is stronger on passive streams because it is a generic changepoint detector. PageHinkley is reliable but slower, KSWIN is too conservative on this signal, and NoDrift is the sanity baseline. In the active setting, however, the relevant anomaly is not merely distributional change but coercive maintenance of an apparently healthy world-model fit. Here CI^E detects masking onset in all 12 runs with median delay 5.0, whereas ADWIN detects only 7 of 12 masking onsets and does so with median delay 279.0. For the collapse transition the same ordering persists: ADWIN detects only 5 of 12 runs, while CI and CI^E detect all of them, with CI^E firing earliest. This is the regime in which the effort-corrected index provides information that a generic passive detector does not encode.

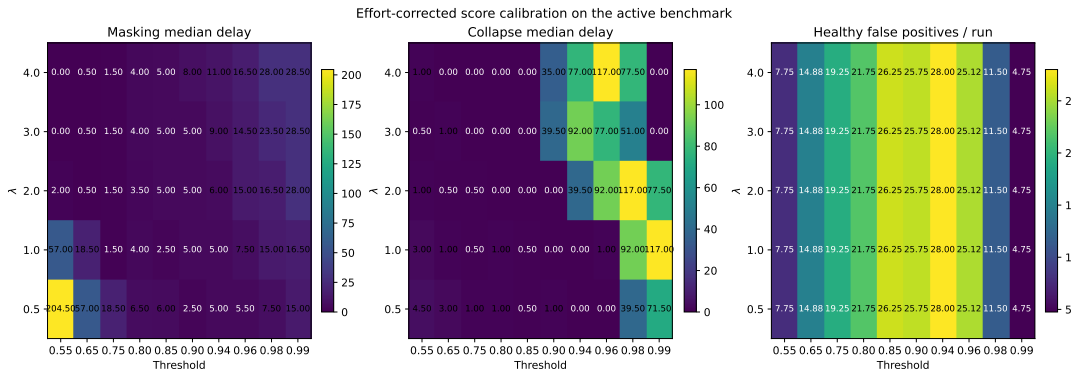


Figure 8: Calibration grid for CI^E over effort penalty λ and threshold. The rate panel is omitted because it saturates across most of the grid; the figure instead reports masking delay, collapse delay, and healthy false positives, which are the informative calibration axes. The default setting ($\lambda = 3.0$, threshold 0.80) remains a low-delay compromise.

4.6 Experiment C: ELEC2 Early-Warning Lead Time

Dataset. The ELEC2 dataset [Harries, 1999] records 45,312 hourly observations of electricity demand in New South Wales, Australia, exhibiting documented concept drift due to market deregulation. We apply a 1D Gaussian tracker to the normalized demand stream.

Protocol. Failures are detected by the Page-Hinkley test [Harries, 1999, threshold = 84, $\delta = 0.01$], yielding 113 regime-shift events on the normalized NSW demand stream. $\hat{C}$ is computed from a 1D Gaussian tracker on the normalized target stream, with all detector hyperparameters fixed before evaluation. The passive benchmark therefore uses no effort proxy and no adaptive threshold retuning on future events. CUSUM, forgetting-factor RLS, and scalar Kalman baselines are run on the same normalized stream, while ADWIN consumes the residual stream used in the original benchmark. The Fréchet baseline compares each current window to the healthy prefix in the same scalar representation space. $\hat{C}$ warnings are recorded on each falling-edge crossing of $\hat{\sigma}_P < 0.295$. Because ELEC2 exposes no direct actuation or effort channel, this benchmark uses the estimable score rather than CI^E ; the effort-corrected score is reserved for the active/coercive setting of section 4.5. Lead time is defined as the gap between a warning and the nearest subsequent Page-Hinkley event, using an adaptive maximum gap set to half the median inter-event interval (nearest-future matching; see appendix A for implementation).

Results. Across the 45,312-step stream, the fixed- $n = 100$ score issues 58 warnings (falling-edge crossings of $\hat{\sigma}_P < 0.295$), of which **18 are matched to a subsequent Page-Hinkley event** within the adaptive matching gap (half the median inter-event interval; 166 steps here). The median lead time is **94 steps**; mean is 96.9; minimum 45; maximum 157. For comparison, ADWIN with $\delta = 0.03$ on the same residual stream issues 114 warnings, of which 54 are matched to a subsequent Page-Hinkley event, at 47.4% precision and median lead 84.5. At lower thresholds the detector becomes more conservative; at higher thresholds it produces more warnings but the same one-to-one matching rule reduces effective precision. On this passive benchmark ADWIN remains the strongest generic change detector under this calibration, while the score continues to serve as a tuneable structural diagnostic whose score semantics carry over to the active setting where CI^E becomes the correct observable.

4.7 Experiment D: Dynamic n_t^* Optimization on ELEC2

Proposition 2.11 provides the self-tuning rule $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$. This experiment tests whether deploying that rule dynamically on ELEC2 improves on the fixed- $n = 100$ score baseline while remaining interpretable next to a strong passive detector.

Protocol. We estimate the local drift rate at each step using an exponentially weighted moving average (EMA, decay $\alpha = 0.05$) of the absolute change in mean between consecutive windows of $d = 50$ steps: $\hat{\zeta}_t = \text{EMA}_\alpha(|\bar{s}_{t-d:t} - \bar{s}_{t-2d:t-d}|/d)$. The constant C_K is calibrated once on the first 2,000 steps (near-stationary): $C_K \approx 1.25 \overline{|s_t - \bar{s}|} \cdot \sqrt{100}$. The resulting $n_t^* = (C_K/\hat{\zeta}_t)^{2/3}$ is clipped to $[30, 300]$, with $\hat{\zeta}_t$

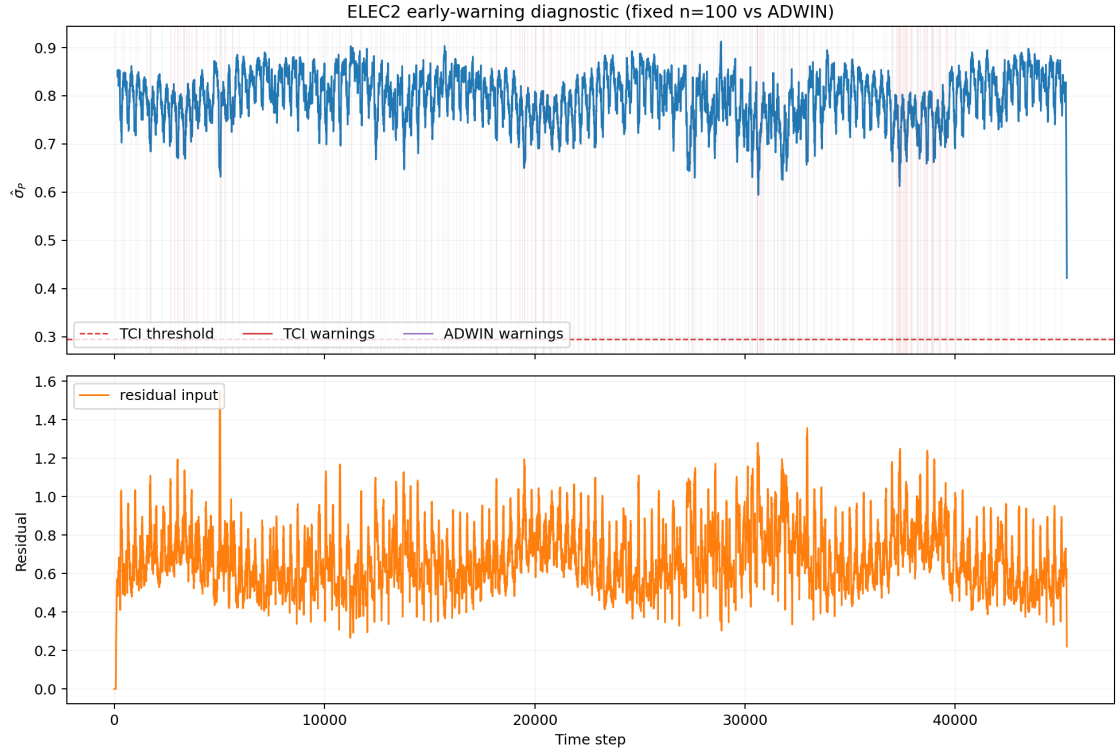


Figure 9: ELEC2 early-warning diagnostic. *Top*: the Coherence Index estimate $\widehat{CI}$ via $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark CI warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom*: residual signal fed to ADWIN (50-step rolling mean).

estimated from consecutive 24-step means and EMA decay $\alpha = 0.03$. The full score sweep in the figure includes fixed $n = 50, 100$, and 300 as references. The main quantitative comparison reported in the table uses the validated fixed- $n = 100$ baseline, the dynamic n_t^* rule, and ADWIN on the same residual stream as an external passive baseline. Warnings and lead times are computed identically to section 4.6, using the same warning threshold 0.295 and one-to-one future matching.

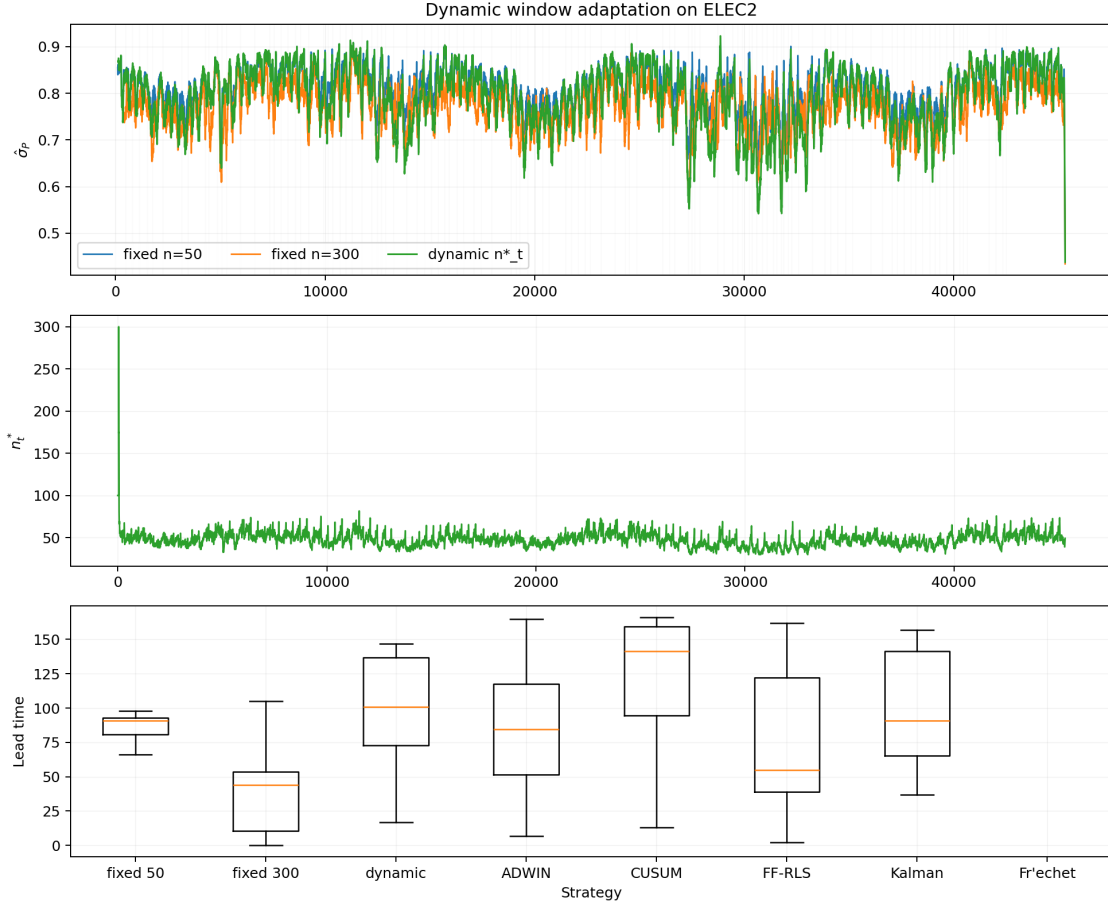


Figure 10: Dynamic n_t^* optimization on ELEC2 ($C_K \approx 1.25 \overline{|s_t - \bar{s}|} \sqrt{100}$, EMA $\alpha = 0.03$, drift window $d = 24$). *Top*: $\hat{\sigma}_P$ (100-step mean) for all three strategies; grey lines are PH regime shifts. *Middle*: dynamic window size n_t^* , contracting sharply near regime shifts and expanding during stable periods. *Bottom*: lead-time distributions, including ADWIN, CUSUM, forgetting-factor RLS, scalar Kalman, and Fréchet baselines.

Results. The results are qualitatively consistent with proposition 2.11, but the passive-stream nature of ELEC2 keeps the advantage modest. Relative to the validated fixed- $n = 100$ baseline, the dynamic rule increases matched leads from 18 to 29 and slightly improves median lead time from 94 to 101, but both remain well behind ADWIN on raw passive detection.

The added passive baselines do not change that ordering. CUSUM is highly sensitive but imprecise; FF-RLS and Kalman sit between the fixed-window score and ADWIN, while the Fréchet baseline is too conservative on ELEC2 to fire at this calibration. The kernel MMD baseline (Gaussian RBF, median-heuristic bandwidth,

Table 8: ELEC2 early-warning comparison: fixed windows, dynamic n_t^* , generic passive baselines, and ADWIN. 113 PH failures; warning level = 0.295; max gap adaptive (half the median inter-event interval; 166 steps here).

Strategy	Warnings	Leads	Precision	Median lead
Fixed $n = 50$	24	11	46%	91
Fixed $n = 100$	58	18	31%	94
Fixed $n = 300$	83	34	41%	44
Dynamic n_t^*	95	29	31%	101
CUSUM	793	96	12%	141.5
FF-RLS	151	45	30%	55
Kalman	96	24	25%	91
Fréchet	0	0	0%	NA
MMD	131	48	37%	74
ADWIN ($\delta = 0.03$)	114	54	47%	84.5

window vs. prefix comparison) achieves 48 matched leads at 37% precision, placing it between ADWIN and the score-based diagnostics. None overtakes ADWIN on matched leads or precision. That ordering is expected: ADWIN is designed for passive changepoint detection, whereas the score targets structural validity under feedback and extends naturally to the effort-corrected CI^E used in the active masking experiment. The Bikes benchmark shows the same passive-stream pattern on a longer real-world demand sequence.

The trade-off is structural: the dynamic rule improves on the static score baseline without changing the basic passive-stream ordering. That is the honest reading carried forward into the later active benchmarks.

The dynamic rule is also moderately robust to calibration error, as predicted by the $2/3$ sensitivity of n_t^* . On ELEC2, scaling $\hat{\zeta}_t$ by $0.75\times$, $1.5\times$, and $2.0\times$ moves the mean dynamic window from 47.7 to 57.8, 36.7, and 31.9, while matched leads change from 29 to 22, 21, and 19. Perturbing C_K by the same factors gives mean windows 39.5, 62.5, and 75.7 with 25, 18, and 19 matched leads. The Bikes benchmark is more tolerant: $\hat{\zeta}_t$ perturbations yield 92, 76, and 74 matched leads, while scaling C_K upward to $1.5\times$ and $2.0\times$ actually increases matched leads to 98 and 110. These shifts are material but not catastrophic, which is the practical point of the sublinear sensitivity law.

The Sinkhorn runtime study makes the regime dependence of C_K more concrete. At the largest window size, increasing ε from 0.02 to 1.0 reduces mean runtime from 4.206 ms to 0.658 ms and reduces mean absolute bias from 1.1115 to 0.1308; changing dimension from $d = 2$ to $d = 256$ shifts the same quantities from 5.410 ms and 0.1077 to 2.346 ms and 1.3918. The right reading is not a closed form for $C_K(d, \varepsilon)$, but a calibrated constant whose value varies with geometry and regularization while the $\zeta^{1/3}$ and $\zeta^{-2/3}$ exponents remain intact.

4.8 Experiment E: Real-Data Masking on KuaiRand

Dataset and protocol. We treat KuaiRand-Pure as the real-data masking benchmark. The random-exposure slice is treated as healthy, the first half of the later

standard-recommendation slice as bubble formation, and the second half as collapse. The benchmark is evaluated on 367 users with at least 20 interactions in each relevant phase. User signals are computed on 20-step windows over `watch_ratio` with video-tag exposure summaries; the effort term is the KL divergence between the current exposure-tag distribution and the user’s healthy random-exposure baseline. The KuaiRand implementation uses a surrogate σ_A based on normalized exposure entropy rather than on replicated action logs, so the benchmark should be read as an operational logged-data proxy for actuation reproducibility rather than as exact $e^{-H(\mathcal{A}|\mathcal{P})}$. To keep the comparison honest, the score and CI^E thresholds are calibrated only on healthy random-exposure windows, using the 20th percentile of the healthy score distribution, and the reported E_0 is the median healthy effort scale. We also report an EWMA-smoothed CI^E baseline to distinguish genuine masking from simple temporal smoothing. The main metric is a user-level detection rate: a user counts as detected for bubble or collapse if the corresponding warning fires within a fixed four-window neighborhood before the annotated phase boundary, and the reported uncertainty intervals are paired bootstrap intervals over users. The companion protocol appendix also records shuffled-label, random-boundary, and weak off-policy sanity controls. These serve as safeguards against trivial exposure-shift explanations, not as causal identification.

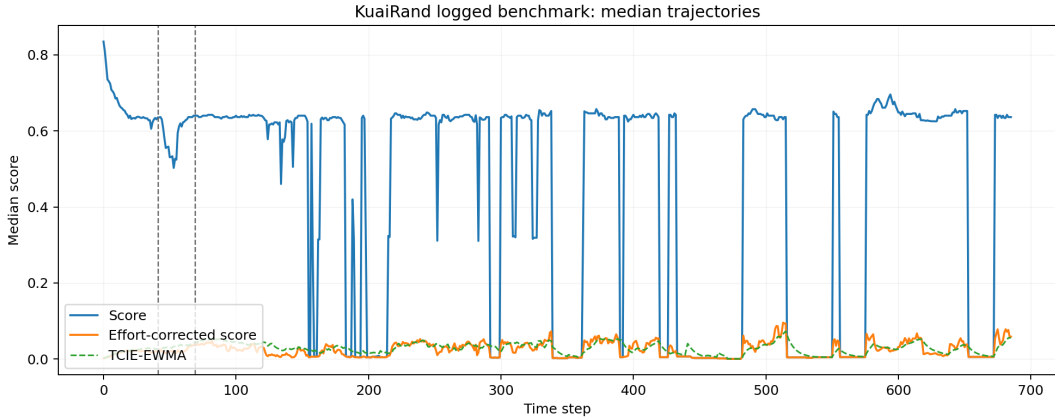


Figure 11: KuaiRand proxy-level logged benchmark. The random-exposure phase is treated as healthy, the first half of the later standard-recommendation slice as bubble formation, and the second half as collapse. Thresholds for CI, CI^E , and EWMA-smoothed CI^E are calibrated only on healthy windows.

Results. Under this healthy-only calibration, the effort-corrected score is the strongest score-based diagnostic in the logged benchmark. Relative to CI, CI^E raises bubble detection from 45.8% to 69.2% and collapse detection from 54.8% to 70.0%, while the user-level paired-bootstrap improvements stay strictly positive: +0.234 for bubble onset with 95% interval [0.183, 0.292], and +0.153 for collapse onset with 95% interval [0.098, 0.207]. The EWMA-smoothed baseline sits in between, as a conservative baseline should: it damps short fluctuations but does not match the raw effort-aware diagnostic. The passive detectors behave unevenly once all of them see the same effort-adjusted signal: KSWIN is the most sensitive generic baseline here, while ADWIN and PageHinkley stay inert under this calibration. For

Table 9: KuaiRand proxy-level logged benchmark summary on 367 users. Under healthy-only calibration, CI^E improves over CI with paired bootstrap deltas $+0.234$ for bubble onset and $+0.153$ for collapse, both with 95% intervals excluding zero.

Phase	Detector	Detections	Rate	Median delay
Bubble onset	CI	168/367	0.458	27.0
Bubble onset	CI^E	254/367	0.692	31.8
Bubble onset	CI^E -EWMA	194/367	0.529	35.5
Bubble onset	ADWIN	0/367	0.000	NA
Bubble onset	PageHinkley	0/367	0.000	NA
Bubble onset	KSWIN	225/367	0.613	23.0
Bubble onset	NoDrift	0/367	0.00	NA
Collapse onset	CI	201/367	0.548	37.0
Collapse onset	CI^E	257/367	0.700	45.0
Collapse onset	CI^E -EWMA	208/367	0.567	41.8
Collapse onset	ADWIN	0/367	0.000	NA
Collapse onset	PageHinkley	0/367	0.000	NA
Collapse onset	KSWIN	357/367	0.973	26.5
Collapse onset	NoDrift	0/367	0.00	NA

fairness, the comparison is run at a fixed operating point: the score-based thresholds are fit on healthy windows only, the reported intervals are paired bootstraps over users, and all generic detectors consume the same effort-adjusted scalar signal rather than detector-specific inputs. This is therefore not an exhaustive ROC analysis. The point is not that generic detectors fail uniformly, but that the effort term exposes a form of control-induced coherence that passive alarms do not encode directly.

The KuaiRand λ sweep gives the same calibration message as the particle tracker. Bubble detection for CI^E rises from 0.463 at $\lambda = 0.5$ to 0.575 at $\lambda = 2.0$, 0.692 at $\lambda = 3.0$, and 0.747 at $\lambda = 4.0$, while healthy false positives rise from 0.777 to 1.215, 1.387, and 1.632 per user. Together with the particle masking-gap sweep, this makes $\lambda \in [2, 3]$ the stable operating region and justifies the default $\lambda = 3$ as a compromise rather than a hand-tuned optimum. The reference scale E_0 behaves as the same calibration logic predicts. A smaller reference scale makes the effort penalty fire more aggressively, raising bubble detection to 0.845 at $E_0 \times 0.5$ but also increasing healthy false positives to 1.837 per user; a larger scale $E_0 \times 2.0$ relaxes the penalty, dropping bubble detection to 0.529 while reducing healthy false positives to 1.019. The follow-up proxy comparison points in the same direction. Replacing the KL effort proxy by total-variation or exposure-Gini alternatives preserves the qualitative ordering $CI^E > CI$ on the logged benchmark, but both alternatives underperform KL at the same healthy-only calibration. We therefore keep KL as the default operational proxy whenever exact action-channel logs are not available.

Taken together, the dynamic-window perturbations, particle λ sweep, KuaiRand λ sweep, and KuaiRand proxy swap form the main sensitivity analysis: the operating point matters, but the central qualitative picture survives the tested perturbations. Figure 12 collects the main calibration axes in one place: the Sinkhorn ε runtime/bias trade-off, the KuaiRand λ and E_0 sweeps, the logged effort-proxy swap, threshold sensitivity under healthy-only calibration, and the particle masking-gap

sweep.

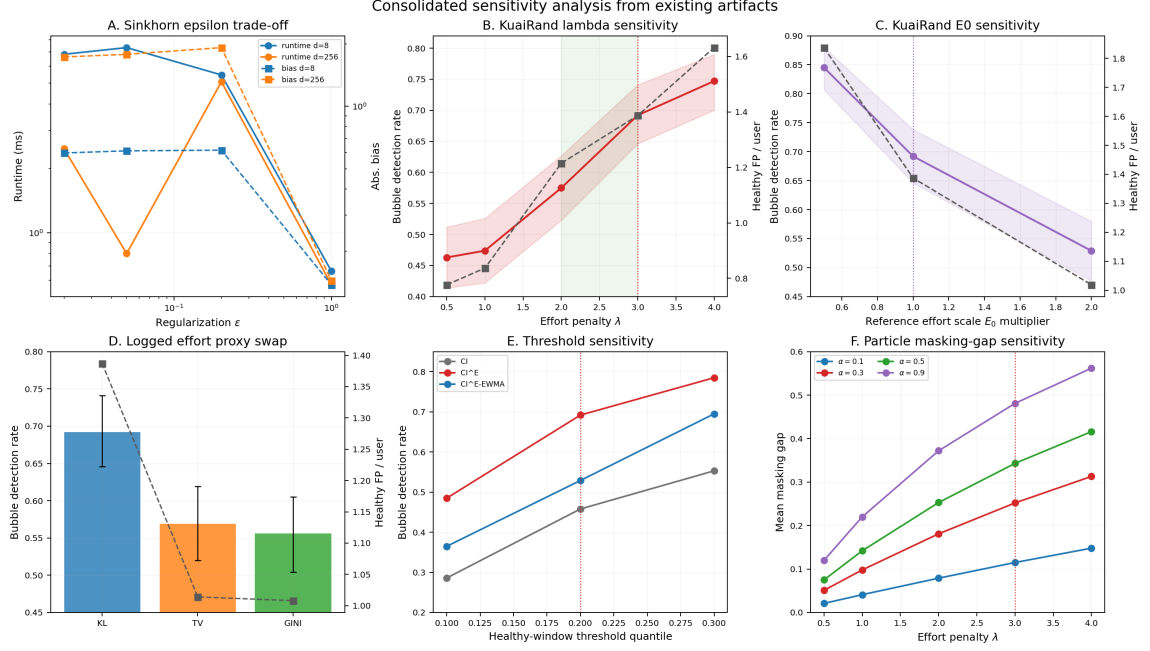


Figure 12: Consolidated sensitivity analysis from existing artifacts. *A*: Sinkhorn regularization ϵ trades runtime against bias, so it should be reported as an operational calibration knob rather than fixed universally. *B–C*: On KuaiRand, stronger effort penalties and smaller reference scales raise bubble sensitivity but also increase healthy false positives, making calibration part of the method rather than a nuisance detail. *D*: KL is the strongest logged effort proxy among the tested options. *E*: The ordering $CI^E > CI^E\text{-EWMA} > CI$ survives threshold-quantile perturbations under healthy-only calibration. *F*: In the particle benchmark, the masking gap grows monotonically with λ and with the action-coupling strength α , confirming that the effort correction is responding to coercive control rather than passive drift alone.

4.9 Experiment G: Bikes Early-Warning Lead Time

Dataset. The River Bikes stream records a long real-world bike-demand sequence with seasonal and weather-driven changes. We normalize the target stream to unit variance and use it as a second passive benchmark beyond ELEC2.

Protocol. As in ELEC2, Page-Hinkley events on the normalized stream define the warning reference. The same fixed-window score strategies ($n = 50, 100, 300$), the dynamic n_t^* rule, and ADWIN on the residual stream are evaluated under the same one-to-one future matching rule. The dynamic rule uses the same deployment recipe as ELEC2: C_K is calibrated on the initial prefix, n_t^* is clipped to a bounded range, and the warning threshold is held fixed rather than tuned against future failures. CUSUM, forgetting-factor RLS, and scalar Kalman baselines are evaluated on the normalized stream with the same thresholding discipline. The Fréchet baseline compares each current window to the healthy prefix in the same representation space.

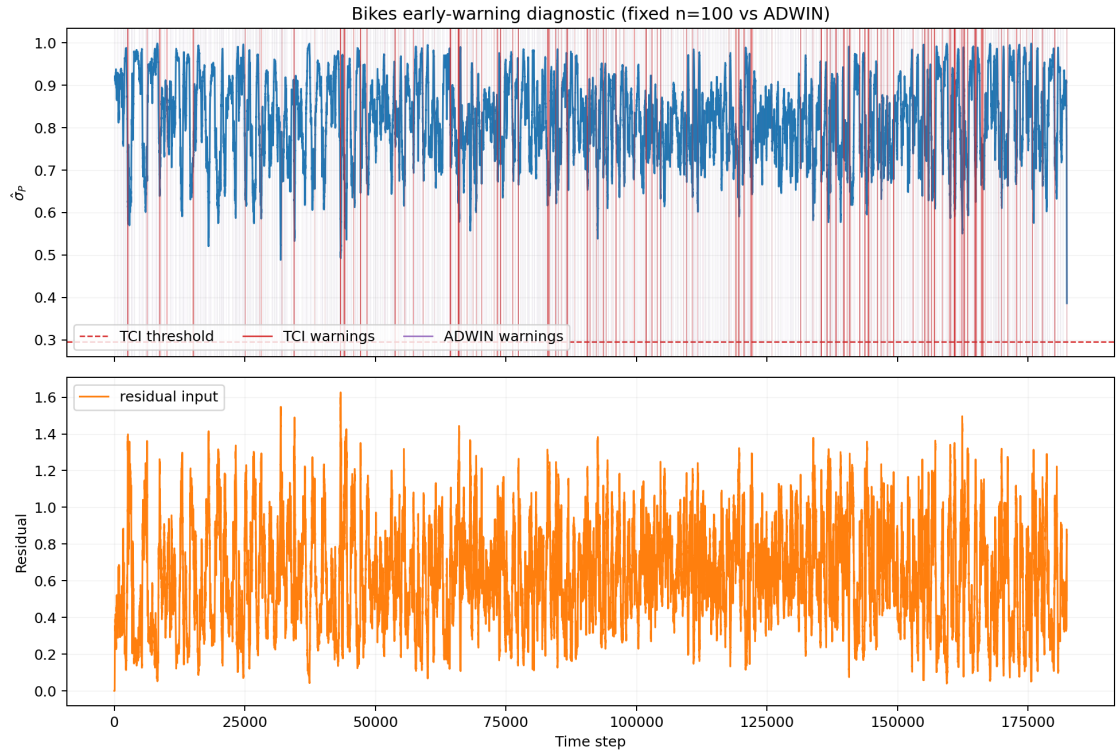


Figure 13: Bikes early-warning diagnostic. *Top*: $\hat{\sigma}_P(t)$ (sliding window $n = 100$) with warning threshold 0.295 (dashed red). Vertical lines mark CI warnings (red), ADWIN warnings (purple), and Page-Hinkley failures (grey). *Bottom*: residual signal fed to ADWIN (50-step rolling mean).

Table 10: Calibration summaries for the dynamic window rule, effort penalty, reference effort scale, and logged-data effort proxies.

Setting	Perturbation / λ	Primary metric	Auxiliary metric
Particle ($\alpha = 0.3$)	$\lambda = 0.5$	$CI^E = 0.898$	$M_t = 0.051$
Particle ($\alpha = 0.3$)	$\lambda = 2.0$	$CI^E = 0.768$	$M_t = 0.181$
Particle ($\alpha = 0.3$)	$\lambda = 3.0$	$CI^E = 0.697$	$M_t = 0.252$
Particle ($\alpha = 0.3$)	$\lambda = 4.0$	$CI^E = 0.636$	$M_t = 0.313$
KuaiRand	$\lambda = 0.5$	bubble rate 0.463	healthy FP/u 0.777
KuaiRand	$\lambda = 2.0$	bubble rate 0.575	healthy FP/u 1.215
KuaiRand	$\lambda = 3.0$	bubble rate 0.692	healthy FP/u 1.387
KuaiRand	$\lambda = 4.0$	bubble rate 0.747	healthy FP/u 1.632
KuaiRand	$E_0 \times 0.5$	bubble rate 0.845	healthy FP/u 1.837
KuaiRand	$E_0 \times 1.0$	bubble rate 0.692	healthy FP/u 1.387
KuaiRand	$E_0 \times 2.0$	bubble rate 0.529	healthy FP/u 1.019
KuaiRand	proxy = KL	bubble rate 0.692	healthy FP/u 1.387
KuaiRand	proxy = TV	bubble rate 0.569	healthy FP/u 1.014
KuaiRand	proxy = Gini	bubble rate 0.556	healthy FP/u 1.008
ELEC2 dyn. n_t^*	baseline	29 matched leads	mean window 47.7
ELEC2 dyn. n_t^*	$\hat{\zeta}_t \times 0.75$	22 matched leads	mean window 57.8
ELEC2 dyn. n_t^*	$\hat{\zeta}_t \times 1.5$	21 matched leads	mean window 36.7
ELEC2 dyn. n_t^*	$C_K \times 0.75$	25 matched leads	mean window 39.5
ELEC2 dyn. n_t^*	$C_K \times 1.5$	18 matched leads	mean window 62.5
Bikes dyn. n_t^*	baseline	85 matched leads	mean window 113.3
Bikes dyn. n_t^*	$\hat{\zeta}_t \times 0.75$	92 matched leads	mean window 136.1
Bikes dyn. n_t^*	$\hat{\zeta}_t \times 1.5$	76 matched leads	mean window 86.9
Bikes dyn. n_t^*	$C_K \times 1.5$	98 matched leads	mean window 146.3
Bikes dyn. n_t^*	$C_K \times 2.0$	110 matched leads	mean window 173.4

Results. On Bikes, ADWIN again leads on matched leads as a generic passive detector. However, the kernel MMD baseline is the strongest newcomer: at 49% precision with 75 matched leads and a median lead time of 22 steps, it provides earlier warnings than any other detector including ADWIN (median 92). The dynamic window rule improves modestly over fixed $n = 100$ in precision but not in lead count. The CUSUM, FF-RLS, Kalman, and Fréchet baselines remain weaker than ADWIN under this calibration, which confirms that the passive-stream caveat is not ELEC2-specific.

4.10 Diagnostic Regime Map

This summary shows which diagnostic is most informative in each regime. Passive streams such as ELEC2 and Bikes still favor generic changepoint detectors; active systems under coercive masking require CI^E ; and logged recommendation data use the same effort-aware logic with a proxy effort signal.

Table 11: Bikes early-warning comparison: fixed windows, dynamic n_t^* , generic passive baselines, and ADWIN. The same one-to-one future matching rule is used as in ELEC2.

Strategy	Warnings	Leads	Precision	Median lead	Min lead
Fixed $n = 50$	572	75	13%	123	0
Fixed $n = 100$	989	101	10%	87	0
Fixed $n = 300$	3,363	137	4%	58	0
Dynamic n_t^*	717	85	12%	127	0
CUSUM	5,928	384	6%	214	11
FF-RLS	15,687	327	2%	220	0
Kalman	13,985	319	2%	220	0
Fréchet	21	4	19%	154.5	65
MMD	152	75	49%	22	0
ADWIN ($\delta = 0.03$)	477	250	52%	92	0

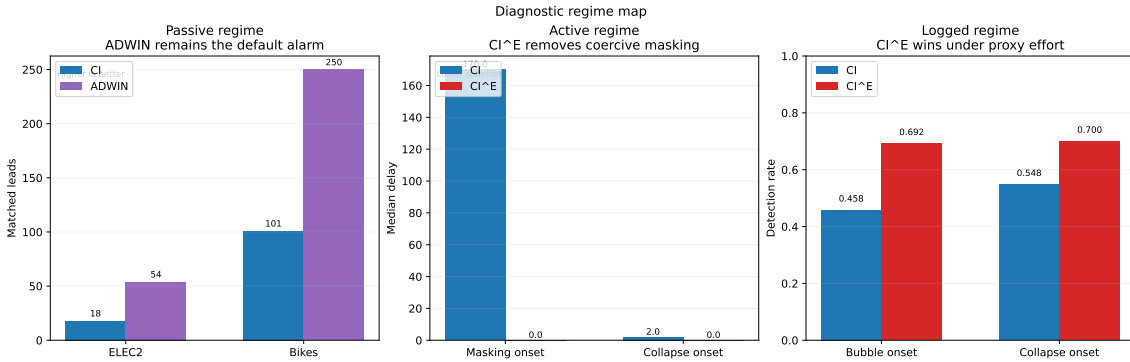


Figure 14: Regime-first diagnostic summary. Passive streams favor generic drift alarms, active feedback exposes the need for CI^E , and logged recommendation data reward proxy-aware effort correction. The figure places each diagnostic in the regime where it is most informative; a median delay of 0.0 means same-timestep detection.

4.11 Passive and Active Diagnostics Are Complementary

The experiments support a clean division of labor. On passive streams such as ELEC2 and Bikes, the relevant anomaly is changepoint detection on exogenous data. In that regime, generic detectors such as ADWIN are the right default tools, and the score should be read as a structural diagnostic with tunable semantics rather than as a universal replacement for passive alarms.

On active or logged benchmarks, the target anomaly changes. In the particle tracker and KuaiRand, the problem is not only that the data distribution moves, but that intervention can keep apparent model-world fit high while a stale model is being protected by effort. In that regime, CI^E and the masking gap are the relevant observables. KSWIN remains a competitive generic detector on KuaiRand, but it pays with false positives comparable to or higher than the effort-aware diagnostic, while ADWIN and PageHinkley contribute little under the same calibration. The practical conclusion is therefore complementary rather than adversarial: use passive drift detectors for passive streams, and use effort-aware coherence when the system can actively shape the observations it is judged against.

4.12 Discussion: Limitations of the Empirical Validation

Four limitations bound the scope of these results. First, the Gaussian-linear setting satisfies H-ISS-P/A/ Φ /Env by construction (example 7.4); non-Gaussian, non-linear systems require case-by-case hypothesis verification. Second, the Sinkhorn rate is validated only for the finite-memory, sliding-window estimator class analyzed here, and only in the variance-dominated regime ($\zeta \leq 0.001$, $n \leq 100$); high-dimensional or fast-drift systems require the debiased S_ϵ with lag correction. Third, the ELEC2 failure definition (Page-Hinkley events) is one operational choice; different failure definitions yield different lead-time statistics. Fourth, the dynamic n_t^* strategy’s precision depends critically on $\hat{\zeta}_t$ estimation quality; a more accurate real-time drift estimator would sharpen the precision-recall frontier.

The Gaussian tracker provides the reproducible, analytically tractable validation of the trade-off and diagnostic claims. High-dimensional, non-Gaussian, and non-linear extensions constitute open directions for future validation (section 11).

5 Mathematical Definitions

We collect the formal setup here after the main diagnostic and empirical claims have been stated. The early sections use only the minimum needed: a representation $\mathcal{P}$, an actuation rule $\mathcal{A}$, and an adaptation operator Φ evolving under drift.

5.1 State Spaces and Information State

Definition 5.1 (State space). Let S be a non-empty measurable space (the *signal space*). The *information state space* $\mathcal{I}$ is the space of all probability measures on S equipped with a metric $d_{\mathcal{I}}$ compatible with weak convergence. Unless otherwise specified, $d_{\mathcal{I}}$ denotes the L^2 -Wasserstein distance W_2 .

Definition 5.2 (Potentiality). The *Potentiality* at time t is a measurable map

$$\mathcal{P}(S, t) \in \mathcal{I},$$

representing the system’s current probability measure over possible world states. $\mathcal{P}$ is ε -coherent at time t if $W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \varepsilon$, where $\mathcal{P}^*(S, t)$ denotes the true marginal distribution of the environment at time t .

Remark 5.3 (Computational realization of W_2). The L^2 -Wasserstein distance W_2 is chosen as the metric on $\mathcal{I}$ for its geometric faithfulness: unlike f -divergences such as KL divergence, W_2 inherits the geometry of the underlying signal space S and behaves well under spatial translations of the distribution support, a desirable property when concept drift shifts the support of $\mathcal{P}^*$. In high-dimensional signal spaces, exact computation of W_2 costs $O(n^3 \log n)$ via network-flow solvers. For streaming applications requiring real-time Φ_t updates, the entropic regularization of Cuturi [2013] replaces W_2 with the Sinkhorn divergence S_ϵ , computable in $O(n^2/\epsilon^2)$ via parallelizable matrix-scaling iterations [Altschuler et al., 2017]. S_ϵ converges to W_2^2 as $\epsilon \rightarrow 0$ and breaks the worst statistical scaling of raw empirical W_2 at the cost of a bias that depends on ϵ and the ambient geometry [Genevay et al., 2019]. The theoretical results hold for W_2 ; their computational realization in streaming contexts is demonstrated in the Gaussian-tracker experiments (section 4; code at appendix A).

Definition 5.4 (Actuation). The *Actuation operator* is a measurable map

$$\mathcal{A}_t: \mathcal{I} \rightarrow \mathcal{O},$$

where $\mathcal{O}$ is the output space (distinct from lowercase o , which denotes a specific output element $o \in \mathcal{O}$). $\mathcal{A}_t$ is δ -reproducible if for any $p \in \mathcal{I}$ and any two executions ω, ω' , $d_{\mathcal{O}}(\mathcal{A}_t(p; \omega), \mathcal{A}_t(p; \omega')) \leq \delta$. A deterministic actuation is 0-reproducible.

Definition 5.5 (Convergence operator). The *Convergence operator* is a measurable map

$$\Phi_t: \mathcal{I} \times \mathcal{O} \times S \rightarrow \mathcal{I},$$

mapping the current information state, the most recent actuation, and an incoming signal to an updated information state. Φ_t is *coherence-restoring* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \geq 0$,

$$W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \beta(W_2(\mathcal{P}(S, 0), \mathcal{P}^*(S, 0)), t) + \gamma\left(\sup_{s \leq t} \|u(s)\|\right),$$

where $u(s)$ represents the environmental drift input at time s .

Definition 5.6 (Three-Role System). An information system $I(S, t)$ is a *three-role system* if it is equipped with all three operators $(\mathcal{P}, \mathcal{A}, \Phi)$ satisfying definitions 5.2, 5.4 and 5.5 respectively. The composite information state is

$$I(S, t) := (\mathcal{P}(S, t), \mathcal{A}_t, \Phi_t) \in \mathcal{I} \times \mathcal{F}_{\mathcal{O}} \times \mathcal{F}_{\mathcal{I}}.$$

5.2 ISS Definitions

Definition 5.7 (Class $\mathcal{K}$, $\mathcal{KL}$). A function $\gamma: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is class $\mathcal{K}$ if it is continuous, strictly increasing, and $\gamma(0) = 0$. A function $\beta: \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}_{\geq 0}$ is class $\mathcal{KL}$ if $\beta(\cdot, t) \in \mathcal{K}$ for each t and $\beta(r, \cdot) \rightarrow 0$ as $t \rightarrow \infty$ for each $r > 0$.

Definition 5.8 (Input-to-State Stability). A dynamical system $\dot{x} = f(x, u)$ is *input-to-state stable* (ISS) if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for every initial state $x(0)$, every essentially bounded input u , and all $t \geq 0$,

$$\|x(t)\| \leq \beta(\|x(0)\|, t) + \gamma(\|u\|_{\infty}).$$

6 Why Action Enters the Update in Closed-Loop Tracking

In closed-loop tracking, the finite-memory law concerns error once a representation is already in place, but action-effective systems face a prior bookkeeping question: what must an update rule remember in order to remain coherent under feedback? In open-loop settings, combining past state and new observations is enough. In closed-loop settings, however, the system's own actions alter future observations. The update therefore needs to keep track of what the system did, not just of what it saw. This point is aligned with standard action-conditioned belief-state semantics; the role here is to make that dependency explicit for the diagnostic decomposition, without claiming a new filtering theorem.

6.1 External Characterization: Drift-Coherence

Definition 6.1 (Drift-Coherence). An information system $I(S, t)$ is *drift-coherent* if there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for all $t \geq 0$ and all environmental inputs with $\text{ess sup}_{s \leq t} \|u(s)\| \leq M < \infty$:

$$W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \leq \beta(W_2(\mathcal{P}(S, 0), \mathcal{P}^*(S, 0)), t) + \gamma(M).$$

The system maintains bounded tracking error under bounded drift.

Definition 6.2 (Action-Effectiveness). The actuation operator $\mathcal{A}_t$ is *action-effective* in environment $\mathcal{X}$ if there exist distinct outputs $o, o' \in \mathcal{O}$ and a state $s \in S$ such that

$$D_{\text{TV}}\left(P(\cdot \mid s_t = s, o) \parallel P(\cdot \mid s_t = s, o')\right) > 0.$$

If this never occurs, the system is *passive*: its actions do not affect the environmental transition law.

6.2 Belief Update With Feedback

Lemma 6.3 (Belief Update Lemma). *Let $I(S, t)$ be a drift-coherent, action-effective information system. Any update map U whose repeated application preserves drift-coherence in this modeling class must depend on three arguments:*

- (i) *the current internal state $p = \mathcal{P}(S, t)$,*
- (ii) *the most recent action $o = \mathcal{A}_t(\mathcal{P})$, and*
- (iii) *the incoming environmental signal s_{t+1} .*

Each argument cannot be dropped without leaving the class of drift-coherent updates considered here.

Proof. If U omits p , the update becomes memoryless and discards the sufficient summary of the past; under temporally structured drift this prevents bounded tracking error. If U omits o , then in an action-effective environment the update cannot distinguish exogenous drift from changes induced by the system's own actuation. This is precisely the credit-assignment problem that appears in POMDP filtering and belief update semantics [Kaelbling et al., 1998, Katsuno and Mendelzon, 1992, Bonanno, 2024]. If U omits s_{t+1} , no new environmental evidence enters the state estimate and the gap between $\mathcal{P}$ and $\mathcal{P}^*$ grows with drift. In each case, the drift-coherence bound fails. $\square$

Remark 6.4 (Passive observer systems). When actions do not influence transitions, the update can drop the actuation argument and the system is genuinely dyadic. The three-role decomposition is therefore motivated only for action-effective settings.

Corollary 6.5 (Ternary Update Structure). *For action-effective tracking systems, any drift-coherent update map has the form*

$$U: \mathcal{I} \times \mathcal{O} \times S \rightarrow \mathcal{I}.$$

Any ostensibly smaller description either hides one of these arguments inside a state variable or fails drift-coherence.

Proof. This is immediate from lemma 6.3. $\square$

Theorem 6.6 (Minimal Functional Roles). *Let $I(S, t)$ be a drift-coherent, action-effective information system. Any functional decomposition that preserves corollary 6.5 in this class must realize three roles: a stored state, an action channel, and an adaptation rule that maps the state, action, and new observation to an updated state. The tuple $(\mathcal{P}, \mathcal{A}, \Phi)$ is the explicit decomposition used here.*

Proof. By corollary 6.5, the update requires all three arguments (p, o, s_{t+1}) . A decomposition that omits any one of the corresponding roles must either re-encode it implicitly inside another component or lose the update structure required for drift-coherence. The explicit realization used here is $\mathcal{P}$ for stored state, $\mathcal{A}$ for the action channel, and Φ for the update map. $\square$

This necessity result does not yet tell us when such a three-role system is stable or diagnostically meaningful. It only shows that, within the present action-effective modeling class, a dyadic state-observation update is too small unless the action term is re-encoded implicitly inside the state. The stability template that follows supplies the corresponding sufficient conditions.

7 A Minimal Stability Template for Closed-Loop Tracking

The necessity argument in section 6 identifies the three inputs a closed-loop update must use. The complementary question is sufficiency: under what conditions does that three-role decomposition support a meaningful diagnostic rather than just a formal one? To make the scores interpretable, we need a stability template under which representation, actuation, and adaptation each admit a Lyapunov-style quantity. We use an ISS-based cascade model for that purpose.

7.1 Hypotheses

The following assumptions are explicit and application-dependent.

(H-ISS-P) The representation subsystem admits an ISS-Lyapunov function $V_P: \mathcal{I} \rightarrow \mathbb{R}_{\geq 0}$.

(H-ISS-A) The actuation subsystem admits an ISS-Lyapunov function V_A .

(H-ISS- Φ) The adaptation subsystem admits an ISS-Lyapunov function V_Φ .

(H-ISS-Env) The environmental drift input is essentially bounded and evolves slowly enough for Φ_t to remain contractive on the induced state trajectory.

Remark 7.1 (Measure-valued state spaces). When $\mathcal{P}(S, t)$ is measure-valued, the natural potential function is $V_P = \frac{1}{2}W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t))^2$. In Gaussian instances this reduces to an ordinary quadratic tracking error; in more general settings the construction follows the Wasserstein gradient-flow viewpoint [Ambrosio et al., 2005, Mironchenko and Prieur, 2020].

7.2 Main Stability Template

Theorem 7.2 (Conditional Borromean Stability Template). *Under hypotheses (H-ISS-P), (H-ISS-A), (H-ISS- Φ), and (H-ISS-Env), the composite system $I(S, t) = (\mathcal{P}, \mathcal{A}, \Phi)$ with feedback structure*

$$\Phi_t \xrightarrow{\text{updates}} \mathcal{P}(S, t) \xrightarrow{\text{drives}} \mathcal{A}_t \xrightarrow{\text{feeds back}} \Phi_t$$

admits the composite Lyapunov functional $V_{\text{total}} := V_P + V_A + V_\Phi$, which is non-increasing in expectation along trajectories for bounded environmental inputs. Removing any one of the three roles destroys this guarantee in general.

Proof. The cascade ($\Phi \rightarrow \mathcal{P} \rightarrow \mathcal{A}$) is ISS by repeated application of standard cascade arguments [Sontag, 1989, 1995, Dashkovskiy et al., 2010]. The subsystem Lyapunov functions combine into a composite functional whose cross terms are absorbed by the ISS gains. If any one of the three roles is removed, at least one of the required input channels disappears: without Φ the state is not corrected, without $\mathcal{P}$ the action has no bounded state input, and without $\mathcal{A}$ the feedback loop to adaptation is broken. In each case the monotone decrease of V_{total} is no longer guaranteed. $\square$

Remark 7.3 (Scope of the stability template). Theorem 7.2 is a conditional stability template, not a universal theorem about all three-role architectures. Its force is constructive in cases where the subsystem Lyapunov functions can be written down and their gains checked directly. The verification is carried out only for the Gaussian tracker. For non-linear or black-box systems, the template should be read as a checklist of sufficient conditions rather than as an already verified property of the architecture itself.

Example 7.4 (Concrete Gaussian instantiation). Let $\mathcal{P}^*(S, t) \sim \mathcal{N}(\mu_t, \Sigma)$ with

$$\mu_t = \mu_0 + \int_0^t \eta(s) ds, \quad \|\eta\|_\infty < M.$$

Let $\mathcal{P}(S, t) = \mathcal{N}(\hat{\mu}_t, \Sigma)$ be the Kalman estimate, $\mathcal{A}_t(\mathcal{P}) = \hat{\mu}_t$, and Φ_t the Kalman update. Then $V_P = W_2(\mathcal{P}, \mathcal{P}^*)^2 = \|\hat{\mu}_t - \mu_t\|^2$, together with the standard actuation and innovation terms, yields a concrete instance satisfying the ISS template [Jazwinski, 1970]. This Gaussian case is a constructive reference instance for theorem 7.2.

With the stability template in place, the remaining question is diagnostic: what kinds of breakdown this decomposition is meant to reveal, and how those breakdowns should appear in the component scores. The failure modes are named next.

8 Diagnostic Failure Modes

The decomposition $(\mathcal{P}, \mathcal{A}, \Phi)$ becomes useful only when it separates failure regimes that matter empirically. The three failure modes that recur throughout the experiments feed directly into the coherence scores. The aim is not to enumerate every possible pathology of an adaptive system, but to isolate the three ways in which representation, actuation, and adaptation can fail in the diagnostic story.

Proposition 8.1 (Failure Mode 1: Static- $\mathcal{P}$ collapse). *Suppose Φ_t ceases to update $\mathcal{P}$, so $\mathcal{P}(S, t) = \mathcal{P}(S, t_0)$ for all $t > t_0$. If the environment undergoes distributional shift $\mathcal{P}^*(S, t) \neq \mathcal{P}^*(S, t_0)$ for large t , then $W_2(\mathcal{P}(S, t), \mathcal{P}^*(S, t)) \rightarrow \infty$ and the expected actuation error $\mathbb{E}[d_{\mathcal{O}}(\mathcal{A}_t(\mathcal{P}), \mathcal{A}_t^*)]$ is unbounded.*

Proof. $\mathcal{P}(S, t) = \mathcal{P}(S, t_0)$ is fixed while $\mathcal{P}^*$ drifts, so $W_2 \rightarrow \infty$ by definition. Lipschitz continuity of $\mathcal{A}_t$ then propagates the divergence to actuation error. $\square$

Remark 8.2 (Interpretation). FM-1 is the stale-model regime: the representation no longer tracks the world, so downstream actions inherit an increasingly obsolete state estimate.

Proposition 8.3 (Failure Mode 2: Non-deterministic- $\mathcal{A}$ divergence). *If $\mathcal{A}_t$ is not δ -reproducible for any $\delta < \infty$, then $H(\mathcal{A}_t \mid \mathcal{P}(S, t)) > 0$ for all t : actuation carries information not determined by the state model. Distributed replicas of the system diverge, and the composite resists auditing and replication.*

Proof. Non- δ -reproducibility implies $\exists \omega \neq \omega'$ with $\mathcal{A}_t(p; \omega) \neq \mathcal{A}_t(p; \omega')$, giving positive conditional entropy. Auditability requires reconstructing actuation from state; this fails when conditional entropy is positive. $\square$

Remark 8.4 (Interpretation). FM-2 is the non-reproducible-action regime: the model may still be accurate, but the realized action can no longer be reconstructed from the represented state, so replicas and audits diverge.

Proposition 8.5 (Failure Mode 3: Absent- Φ monopolization). *Let $\Phi_t \equiv \text{id}$ (no update). In any closed-loop deployment, actuation feedback shifts $\mathcal{P}^*$; since $\mathcal{P}$ cannot track, the distributional mismatch grows at a rate lower-bounded by the rate of environmental response to actuation.*

Proof. The closed-loop case follows from $\mathcal{P}^*(t+1) \leftarrow f(\mathcal{P}^*(t), \mathcal{A}_t(\mathcal{P}(t)))$ with $\mathcal{P}$ fixed, so the gap $W_2(\mathcal{P}, \mathcal{P}^*)$ grows at least as fast as the variation of f in its second argument whenever $\mathcal{A}_t(\mathcal{P}) \neq \mathcal{A}_t^*$. $\square$

Remark 8.6 (Interpretation). FM-3 is the missing-adaptation regime. In passive settings it appears as a failure to track external drift; in closed-loop settings it can amplify the system's own feedback and produce self-reinforcing misalignment.

These failure modes motivate the score construction that follows. Once the three ways of failing are separated, the natural diagnostic question is how to summarize them in a single quantity without losing the identity of the active bottleneck.

9 Diagnostic Characterization

A structural asymmetry in the framework deserves explicit statement.

Drift-coherence implies the need for all three roles. By lemma 6.3 and theorem 6.6, any drift-coherent, action-effective system must implement the three functional roles $(\mathcal{P}, \mathcal{A}, \Phi)$. Drift-coherence is the stronger behavioral condition; the decomposition is its architectural consequence in action-effective settings.

Implementing the decomposition is not enough. A system may implement $(\mathcal{P}, \mathcal{A}, \Phi)$ structurally while failing to satisfy the ISS hypotheses (H-ISS-P/A/ Φ /Env) required for theorem 7.2. The components may be present but improperly tuned, have inadequate bandwidth, or lack sufficient expressiveness.

Proposition 9.1 (The Score Characterizes Component-Level Failure). *Let $I(S, t)$ be a three-role system under hypotheses $(H\text{-ISS-}P/A/\Phi/Env)$. Then:*

- (i) $\mathcal{C}(I(S, t), t) = 1$ if and only if $V_P(t) = 0$, $H(\mathcal{A}_t \mid \mathcal{P}) = 0$, and $V_\Phi(t) = 0$ simultaneously.
- (ii) $\mathcal{C}(I(S, t), t) \rightarrow 0$ if and only if at least one of the three failure modes is active: $V_P(t) \rightarrow \infty$ (FM-1), $H(\mathcal{A}_t \mid \mathcal{P}) \rightarrow \infty$ (FM-2), or $V_\Phi(t) \rightarrow \infty$ (FM-3).
- (iii) The identity of $\arg \min(\sigma_P, \sigma_A, \sigma_\Phi)$ names the bottleneck component.

Proof. (i) Each score is a strictly decreasing function of its Lyapunov functional or conditional entropy: $\sigma_P = 1 \Leftrightarrow V_P = 0$, $\sigma_A = 1 \Leftrightarrow H(\mathcal{A}_t \mid \mathcal{P}) = 0$, $\sigma_\Phi = 1 \Leftrightarrow V_\Phi = 0$. The min equals 1 iff all three equal 1 simultaneously.

(ii) As $V_P(t) \rightarrow \infty$, $\sigma_P(t) \rightarrow 0$, so $\mathcal{C} \rightarrow 0$; similarly for $V_\Phi(t) \rightarrow \infty$. As $H(\mathcal{A}_t \mid \mathcal{P}) \rightarrow \infty$, $\sigma_A(t) = e^{-H} \rightarrow 0$, so $\mathcal{C} \rightarrow 0$. The converse holds because $\mathcal{C} \rightarrow 0$ requires at least one $\sigma_i \rightarrow 0$, which by the preceding argument requires the corresponding functional to diverge.

(iii) $\mathcal{C} = \min_i \sigma_i$; the argmin identifies which subsystem's functional is largest, i.e., which component is furthest from its reference state. $\square$

Proposition 9.2 (Null-State False Alarm Control for the Gaussian Tracker). *Consider the effort-corrected score $\mathcal{C}^E(I(S, t), t)$ evaluated on the linear-Gaussian tracking environment introduced in section 4.2. Under the null hypothesis of healthy tracking without coercive masking ($\zeta_t = \zeta$, action exactly targets the latent state), the variance of the optimal prediction error converges to a steady-state value P_∞ . At steady state, the false positive rate for a warning threshold $\tau \in (0, 1)$ is strictly controlled by:*

$$\mathbb{P}(\mathcal{C}^E(I(S, t), t) < \tau) \leq 2\Phi^c\left(\sqrt{\frac{2}{P_\infty}}\left(\frac{1}{\tau} - 1\right)\right)$$

where Φ^c is the complementary CDF of the standard normal distribution.

Proof. By definition, $\mathcal{C}^E(t) = \min(\sigma_P(t), \sigma_A^E(t), \sigma_\Phi(t))$. Under the optimal tracking null hypothesis without coercive masking, the environment acts on the true target without excess correction effort ($\Delta_A = 0 \implies \sigma_A^E = 1$) and the variance estimator is perfectly calibrated ($V_\Phi = 0 \implies \sigma_\Phi = 1$). The index thus reduces exactly to the prediction component $\sigma_P(t) = (1 + V_P(t))^{-1}$, where $V_P(t) = \frac{1}{2}(\hat{\mu}_t - \mu_t)^2$.

In the steady state of the scalar Kalman filter, the tracking error $\hat{\mu}_t - \mu_t$ is distributed as $\mathcal{N}(0, P_\infty)$. Consequently, $V_P(t)$ is distributed as $\frac{P_\infty}{2}\chi_1^2$. The false-alarm event $\mathcal{C}^E(t) < \tau$ reduces to $V_P(t) > \frac{1}{\tau} - 1$. Taking the square root, this evaluates directly to the two-sided normal tail probability $2\Phi^c\left(\sqrt{\frac{2}{P_\infty}}\left(\frac{1}{\tau} - 1\right)\right)$, establishing the bound. $\square$

10 Related Work

Tracking under non-stationarity. This work sits at the intersection of three literatures that are usually kept apart: concept drift detection, finite-memory tracking, and action-coupled prediction. In the concept-drift literature, classical detectors

such as ADWIN, DDM, and Page-Hinkley identify changepoints or sustained mean shifts in a monitored signal [Gama et al., 2014, Bifet and Gavaldà, 2007, Gama et al., 2004, Harries, 1999]. Those methods remain the right passive baselines, and the empirical ordering here leaves that conclusion intact: on passive streams they are still the stronger default alarms. The role of the Coherence Index (CI) is narrower. It is a structural diagnostic tied to model fidelity, actuation reproducibility, and update quality, not a universal replacement for generic changepoint detection.

The second neighborhood is nonstationary optimization, adaptive filtering, and online least-squares. There the same cube-root tension between variation and memory already appears, but in Euclidean loss, bandit, dynamic-regret, or parameter-tracking settings rather than in a distributional W_2 /Sinkhorn geometry [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The $1/3$ exponent itself is therefore not presented as wholly new. The more specific claim is that, in this setting, the same balance can be written as an explicit finite-memory error law for causal averaging estimators under W_2 -Lipschitz drift, carried through to an operational selector for window size or forgetting factor, and matched by a minimax obstruction for arbitrary window-restricted estimators on the same drift class. Classical smoothers such as EWMA, forgetting-factor RLS, and Kalman filtering belong to the same broad family of variance-versus-responsiveness trade-offs, but they are usually studied as estimators rather than as components of a closed-loop diagnostic argument. Disturbance-observer work on KF-DOB makes a closely related bias-variance trade-off explicit, but in a model-mismatch setting rather than a drift-controlled memory law [Li et al., 2024]. The EWMA results matter here because they show that the cube-root law is a finite-memory phenomenon rather than a sliding-window artifact. What is added is the combination of that law with a closed-loop diagnostic and a practical memory selector.

Transport-based diagnostics. WATCH uses Wasserstein geometry to detect distribution shift in feature space [Faber et al., 2022b,a]. The relation is close, but the role of transport is different. There, Wasserstein distance is the detector. Here, W_2 and its debiased Sinkhorn surrogate are built into the tracking model itself: they quantify representation error, induce the lag-variance trade-off, and anchor the coherence diagnostics. The point is not simply that OT is a good shift statistic, but that in this setting it belongs to the geometry of what it means for a finite-memory tracker to stay current. On the statistical side, the finite-sample convergence rates of empirical Wasserstein distances depend on both the sample size and the intrinsic dimension of the support [Niles-Weed and Bach, 2019]; these rates inform the constant C_K in our bounds and suggest that practitioners tuning ε in higher dimensions should consult dimension-aware calibration rather than using a fixed regularization schedule. Recent limit theorems for the Sinkhorn divergence also clarify the calibration split that matters here: bootstrap-friendly alternative-regime behavior and a different null limit order for threshold setting [Goldfeld et al., 2024]. In the Gaussian case, exact entropic OT structure is even available in closed form, which helps explain why the prototype runtime study behaves cleanly on Gaussian clouds even when the constants still move with geometry [Akyildiz et al., 2024].

Kernel-based shift detection. Kernel two-sample tests based on the Maximum Mean Discrepancy (MMD) [Gretton et al., 2012] provide a complementary nonpara-

metric approach to distribution shift detection, and several streaming drift detectors build on them. The MMD and W_2 address related but distinct aspects of distributional change: MMD measures discrepancy in a reproducing kernel Hilbert space and is straightforward to threshold for hypothesis testing, while W_2 inherits the metric geometry of the signal space and connects directly to the lag–variance trade-off studied here. We now include an MMD-based detector on the real-world passive streams, so direct empirical comparison is available there; however, the CI/CI^E diagnostics operate at a different level than a shift alarm, since they score the structural bottleneck of a three-role system rather than testing whether two windows are drawn from the same distribution.

Filtering and closed-loop control. The representation–action–adaptation decomposition is motivated by standard state-estimation and control logic, not by a claim that every adaptive system must have three roles. ISS theory supplies the stability template [Sontag, 1989, Dashkovskiy et al., 2010, Mironchenko and Prieur, 2020], while classical filtering provides the concrete Gaussian realization used here [Jazwinski, 1970]. Once actions affect future observations, state update must solve a credit-assignment problem rather than a passive filtering problem.

Belief update semantics. The distinction between revision and update in dynamic worlds [Katsuno and Mendelzon, 1992] gives a formal language for the same issue. Bonanno [2024] makes explicit that action-conditioned belief update must distinguish exogenous change from state changes caused by the agent itself. That observation matters here because it justifies the presence of the action argument in the closed-loop update.

Algorithmic regulation and compression. Ruffini [2026] interprets good regulation through algorithmic compression of realized trajectories. The present framework adds a temporal constraint absent from that picture: even a useful compressed model becomes stale under drift. The cube-root law therefore complements the algorithmic-regulation viewpoint by quantifying how finite memory limits the duration for which compression remains valid in a changing environment.

Action-coupled prediction. The coercive masking regime is related to broader work on systems that change the data-generating process they seek to model. The claim is narrower than a general performative-prediction analysis: the focus here is a diagnostic failure mode in which low observed error can be sustained by intervention, along with an effort-aware correction for that case. This connects most directly to performative prediction and feedback-loop analyses in recommendation and dataset construction, including [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023, Farina and Perdomo, 2026, Taori and Hashimoto, 2023, Pagan et al., 2023, Krauth et al., 2022, Eilat and Rosenfeld, 2023]. The main distinction is that the aim is not to solve the full causal problem of performative feedback or recommendation confounding, but to diagnose when intervention effort masks stale models in active or logged systems. Strategic classification [Hardt et al., 2016] studies a closely related phenomenon from the opposite side: agents modify their features to game a deployed classifier, creating a feedback loop between the model and the data it evaluates. Our

coercive masking regime is the system-side analogue: the system modifies the environment rather than the agents modifying themselves. CI^E does not solve the full strategic or performative problem; it provides a diagnostic for when low observed error should stop counting as evidence of good adaptation. Performative stability asks where retraining dynamics settle under the distribution induced by their own predictions [Perdomo et al., 2020]; recent online-learning results show that broad no-regret procedures can converge to mixed performatively stable equilibria even without strong structural restrictions [Farina and Perdomo, 2026]. Coercive masking asks a complementary question. A system may remain near such an apparent equilibrium while paying a growing intervention cost to hold the world there against exogenous drift. CI^E is designed to measure that maintenance burden directly. In that sense, performative prediction characterizes where the system settles, whereas the diagnostic developed here characterizes how much forcing is required to stay there.

11 Limitations and Open Directions

Finite-memory scope. The cube-root law of proposition 2.11 is exact for the finite-memory averaging class analyzed here, and proposition 2.15 extends the lower-bound side to arbitrary measurable estimators that only see the last m samples of a ζ -Lipschitz drift path. The cube-root exponent therefore survives beyond linear averaging in the critical-window regime, but this still falls short of a universal minimax result over arbitrary recursive or extrapolative adaptive estimators.

Geometry and observability. The present formulation uses W_2 and its debiased Sinkhorn approximation to measure model drift. Extending the same trade-off cleanly to other geometries, such as f -divergences or kernel MMD, remains open. Likewise, CI^E (written formally as $\mathcal{C}^E(I(S, t), t)$ when needed) requires an observable proxy for intervention effort, and the preferred proxy is domain-dependent. This is especially relevant in performative or feedback-loop settings, where intervention and exposure are partially confounded [Perdomo et al., 2020, Hardt and Mendler-D"unner, 2023, Taori and Hashimoto, 2023, Krauth et al., 2022].

Benchmark dependence. The passive benchmarks use Page-Hinkley events as the operational reference for lead time, and different failure definitions would change the resulting alarm statistics. On active and logged benchmarks, the usefulness of CI^E still depends on calibration choices such as the effort penalty λ , the warning threshold, and the local drift estimator $\hat{\zeta}_t$.

Effort proxies and σ_A surrogates. The practical hierarchy of effort proxies is strongest in simulation, weaker in direct observation, and weakest in logged recommendation data. Likewise, σ_A is exact only in settings with replicated or fully specified action channels. In the Gaussian tracker it is constructive, in the particle tracker it is a controlled surrogate based on actuation gap, and in KuaiRand it is a normalized exposure-entropy proxy rather than a direct estimate of $e^{-H(\mathcal{A}|\mathcal{P})}$. The logged benchmark should therefore be read as operational evidence for effort-aware diagnostics, not as full causal identification of intervention dynamics. A stronger

causal claim would require either replica-policy or shadow-policy logs, or an external human-grounded outcome surface against which masking could be falsified rather than inferred from the benchmark’s own internal signals. More richly logged recommender traces that preserve full slate or action-channel realizations would also strengthen the estimand itself: in that regime σ_A could move from a proxy-level surrogate toward the identified tier of the hierarchy in definition 2.5.

Scaling beyond present validation. The Gaussian and particle trackers validate the theory in controlled Gaussian and non-Gaussian settings, and the real-data benchmarks show that the diagnostics remain useful outside the synthetic environment. High-dimensional state spaces, richer action channels, and direct measurements of intervention effort are the next natural tests of the framework. A particularly sharp next benchmark is RLHF-style post-training. That setting instantiates all three roles unusually cleanly: the current policy plays the role of $\mathcal{P}$, sampled completions or actions play the role of $\mathcal{A}$, and the policy update rule plays the role of Φ . Policy-reference KL is already logged as an intervention burden in standard RLHF pipelines [Stiennon et al., 2020, Bai et al., 2022], making it a much cleaner effort proxy than the exposure surrogates available in logged recommendation data. At the same time, reward-model or proxy-reward improvement can be compared against held-out or human-grounded quality, creating an external masking surface that the present KuaiRand benchmark does not have. RLHF is therefore a natural next test bed for the masking gap $M_t = CI - CI^E$ (formally, $\mathcal{C}(I(S, t), t) - \mathcal{C}^E(I(S, t), t)$): the system may maintain high proxy coherence while paying increasing KL cost and drifting away from the external target. Practical simulation frameworks such as AlpacaFarm [Dubois et al., 2024] make that direction especially plausible without requiring full-scale LLM training in the loop.

Opaque adaptation maps. For black-box Φ_t , the framework provides only the template requirement that an ISS-Lyapunov functional V_Φ exist. A CEGIS-style recovery route is not implemented or validated in this repository, so diagnosing σ_Φ for opaque adaptation subsystems remains an open engineering problem.

Computational profile. The Sinkhorn runtime study is a prototype-level NumPy benchmark on Gaussian clouds. It is useful for separating statistical accuracy from wall-clock cost, but it does not claim a production implementation ceiling. The current sweep reaches $d = 256$ and $\varepsilon = 0.02$, and the main practical message is that fixed regularization is not portable across geometry: larger ε is consistently cheaper and more accurate on this grid, while the residual bias and iteration count still depend on dimension. The current calibration is therefore operational rather than asymptotically exact: it is meant to compare tracked drifted clouds at a chosen operating point, not to claim a null-optimal threshold rule. Recent Sinkhorn limit theorems make that split explicit, with distinct alternative-regime and null fluctuation orders [Goldfeld et al., 2024].

Missing adaptive baselines. The passive comparisons emphasize ADWIN, Page-Hinkley, KSWIN, and the NoDrift sanity baseline, and now also include CUSUM, forgetting-factor recursive least squares, scalar Kalman predictors, and a kernel MMD two-sample test (Gaussian RBF, median-heuristic bandwidth) on the

real-world streams. That does not exhaust the baseline space: multivariate MMD detectors, representation-space shift tests, and additional online adaptive learners such as explicit H_∞ filters remain future work, and the present calibration should still be read as benchmark-specific rather than universal.

12 Conclusion

Under drift, memory is not only a resource; it is also a source of staleness. Under feedback, good observed fit is not yet evidence that the model is current, because intervention can keep the world near a stale estimate. Taken together, those two facts point to two claims: a finite-memory tracking floor, and an intervention-aware diagnostic for when apparent coherence is being purchased rather than earned.

The tracking result is a sharp coherence-delay limit for the finite-memory averaging class. When total error is written as a sum of estimation variance and drift-induced staleness, optimizing over memory yields the cube-root law $\mathcal{E}_{\min} = \frac{3}{2} C_K^{2/3} \zeta^{1/3}$ for the estimator class analyzed here. The broader lower-bound argument then shows that the same exponent is not merely a sliding-window artifact: it reappears for arbitrary estimators restricted to the last m samples on a ζ -Lipschitz drift class in the critical-window regime. The practical lesson is blunt: more data can still make a tracker worse if the world moves faster than the memory can stay current.

The diagnostic result addresses a specific failure mode of action-coupled systems: apparent coherence maintained by forcing rather than by adaptation. The decomposition into representation, actuation, and adaptation roles $(\mathcal{P}, \mathcal{A}, \Phi)$ isolates that regime from ordinary degradation, and the Effort-Adjusted Coherence Index (CI^E) is built for exactly that case. The empirical results support the distinction. The Gaussian tracker recovers the expected U-curve and the $\zeta^{1/3}$ scaling law for both sliding windows and EWMA. The particle benchmark shows that feedback masking can keep the standard score high even when the model is stale. On passive streams, generic changepoint detectors such as ADWIN remain the stronger default alarms. On active and logged benchmarks, however, the effort-aware score becomes the right diagnostic because the failure mode is no longer passive drift alone. On KuaiRand, the gain of CI^E over CI is statistically stable under paired bootstrap resampling, but the logged benchmark remains proxy-level evidence rather than causal identification of masking; within that proxy regime, KL is also the strongest effort proxy among the logged alternatives tested here. The practical tuning rule is correspondingly simple: set E_0 to the healthy effort scale, keep λ in the empirically stable 2–3 range, and use the KL proxy when only logged exposure summaries are available. Those settings are not universal optima; they are the operating point supported by the current logged benchmark. The most natural next test of the same idea is RLHF-style post-training, where policy KL already measures intervention burden and proxy-reward divergence provides a sharper external masking surface.

The broader claim is modest, but useful. Beyond the specific results, this framework points toward finite-memory design rules under drift, intrinsic-dimension-aware OT estimation, and principled diagnostics for performative or action-coupled settings. Adaptive systems should not be judged only by how well they fit what they currently see. They should also be judged by how much temporal staleness they can tolerate and by how much intervention is required to preserve the appearance of coherence. Within the scope studied here, the cube-root law explains the tracking

floor, and CI^E explains why low observed error can still be the wrong comfort signal under feedback.

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References

- O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Míguez. Gaussian entropic optimal transport: Schrödinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.
- Jason Altschuler, Jonathan Weed, and Philippe Rigollet. Near-linear time approximation algorithms for optimal transport via Sinkhorn iteration. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 1964–1974, 2017.
- Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2005. doi: 10.1007/978-3-7643-8722-8.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova Das-Sarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. Training a helpful and harmless assistant with reinforcement learning from human feedback. *CoRR*, abs/2204.05862, 2022. doi: 10.48550/arXiv.2204.05862.
- Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.
- Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007. doi: 10.1137/1.9781611972771.42.
- Giacomo Bonanno. A Kripke-Lewis semantics for belief update and belief revision. *Artificial Intelligence*, 328:104065, 2024. doi: 10.1016/j.artint.2023.104065.
- Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL <https://proceedings.mlr.press/v89/cheung19b.html>.

- Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems 26 (NeurIPS)*, pages 2292–2300, 2013.
- Sergey N. Dashkovskiy, Björn S. Ruffner, and Fabian R. Wirth. Small gain theorems for large scale systems and construction of ISS lyapunov functions. *SIAM Journal on Control and Optimization*, 48(6):4089–4118, 2010. doi: 10.1137/090746483.
- Yann Dubois, Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. AlpacaFarm: A simulation framework for methods that learn from human feedback. *CoRR*, abs/2305.14387, 2024. doi: 10.48550/arXiv.2305.14387.
- Itay Eilat and Nir Rosenfeld. Performative recommendation: Diversifying content via strategic incentives. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 9082–9103. PMLR, 2023. URL <https://proceedings.mlr.press/v202/eilat23a.html>.
- Kamil Faber, Roberto Corizzo, Bartłomiej Snieżyński, Michael Baron, and Nathalie Japkowicz. LIFEWATCH: Lifelong wasserstein change point detection. In *International Joint Conference on Neural Networks*, pages 1–8. IEEE, 2022a. doi: 10.1109/IJCNN55064.2022.9892891.
- Malte Faber, Magda Sakr, and Annette Bauer. WATCH: Wasserstein change point detection for high-dimensional time series data. In *2022 IEEE International Conference on Big Data*, pages 3268–3277, 2022b. doi: 10.1109/BigData55660.2022.10020525.
- Gabriele Farina and Juan Carlos Perdomo. The stability of online algorithms in performative prediction. *CoRR*, abs/2602.24207, 2026. doi: 10.48550/arXiv.2602.24207.
- João Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Advances in Artificial Intelligence — SBIA 2004*, volume 3171 of *Lecture Notes in Computer Science*, pages 286–295. Springer, 2004. doi: 10.1007/978-3-540-28645-5_29.
- João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):44:1–44:37, 2014. doi: 10.1145/2523813.
- Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.
- Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.

- Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012. URL <http://jmlr.org/papers/v13/gretton12a.html>.
- Moritz Hardt and Celestine Mendler-Dünnier. Performative prediction: Past and future. *CoRR*, abs/2310.16608, 2023. doi: 10.48550/arXiv.2310.16608.
- Moritz Hardt, Nimrod Megiddo, Christos H. Papadimitriou, and Mary Wootters. Strategic classification. In *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS)*, pages 111–122. ACM, 2016. doi: 10.1145/2840728.2840730.
- Michael Harries. SPLICE-2 comparative evaluation: Electricity pricing. Technical report, University of New South Wales, 1999. Available at <https://www.openml.org/d/151>.
- Andrew H. Jazwinski. *Stochastic Processes and Filtering Theory*, volume 64 of *Mathematics in Science and Engineering*. Academic Press, New York, 1970.
- Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998. doi: 10.1016/S0004-3702(98)00023-X.
- Hirofumi Katsuno and Alberto O. Mendelzon. On the difference between updating a knowledge base and revising it. In Peter Gärdenfors, editor, *Belief Revision*, volume 29 of *Cambridge Tracts in Theoretical Computer Science*, pages 183–203. Cambridge University Press, 1992.
- Karl Krauth, Yixin Wang, and Michael I. Jordan. Breaking feedback loops in recommender systems with causal inference. *CoRR*, abs/2207.01616, 2022. doi: 10.48550/arXiv.2207.01616.
- Shilei Li, Dawei Shi, Xiaoxu Lyu, Jiawei Tang, and Ling Shi. Bias-variance trade-off in kalman filter-based disturbance observers. *CoRR*, abs/2410.05061, 2024. doi: 10.48550/arXiv.2410.05061.
- Lennart Ljung. *System Identification: Theory for the User*. Prentice Hall PTR, Upper Saddle River, NJ, 2 edition, 1999.
- Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL <https://proceedings.mlr.press/v202/mazzetto23a.html>.
- Andrii Mironchenko and Christophe Prieur. Input-to-state stability of infinite-dimensional systems: Recent results and open questions. *SIAM Review*, 62(3): 529–614, 2020. doi: 10.1137/19M1291248.
- Jonathan Niles-Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A): 2620–2648, 2019. doi: 10.3150/18-BEJ1077.

- Nicolò Pagan, Joachim Baumann, Ezzat Elokda, Giulia De Pasquale, Saverio Bolognani, and Anikó Hannák. A classification of feedback loops and their relation to biases in automated decision-making systems. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 7:1–7:14. ACM, 2023. doi: 10.1145/3617694.3623227.
- Juan Perdomo, Tijana Zrnic, Celestine Mendler-D"unner, and Moritz Hardt. Performative prediction. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7599–7609. PMLR, 2020. URL <https://proceedings.mlr.press/v119/perdomo20a.html>.
- Giulio Ruffini. The algorithmic regulator. *Entropy*, 28(3):257, 2026. doi: 10.3390/e28030257.
- Eduardo D. Sontag. Smooth stabilization implies coprime factorization. *IEEE Transactions on Automatic Control*, 34(4):435–443, 1989. doi: 10.1109/9.28018.
- Eduardo D. Sontag. On the input-to-state stability property. *European Journal of Control*, 1(1):24–36, 1995. doi: 10.1016/S0947-3580(95)70005-X.
- Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems*, volume 33, pages 3008–3021, 2020. URL <https://arxiv.org/abs/2009.01325>.
- Rohan Taori and Tatsunori Hashimoto. Data feedback loops: Model-driven amplification of dataset biases. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 33883–33920. PMLR, 2023. URL <https://proceedings.mlr.press/v202/taori23a.html>.
- Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.

A Code Repository

All implementations are publicly available at:

<https://github.com/jovemexausto/coherence-delay-tradeoff>

The repository contains:

- `experiments/run.py` — single entry point for all experiment domains;
- `experiments/gaussian/` and `experiments/particle/` — core experiment logic;
- `experiments/bikes/`, `experiments/elec2/`, and `experiments/kuairand/` — domain-specific benchmarks;

- `REPRODUCIBILITY.md` — setup and run instructions.

All experiments use `uv` for dependency management (`uv sync`) and are reproducible with fixed random seeds as specified in each script.

B KuaiRand Logged Benchmark Protocol

This protocol instantiates the real-data masking benchmark on top of the KuaiRand sequential recommendation logs. The goal is to test whether the Effort-Adjusted Coherence Index (CI^E) can surface exposure concentration and filter-bubble-like drift more clearly than passive detectors and a smoothed-score baseline.

Data and signals

The primary state signal is `watch_ratio`. Auxiliary feedback signals include `click` and `like`; `dwell`, `follow`, and `share` are reserved for secondary validation when available. For user u at time t , let $P_{u,t}$ denote the inferred preference state, $A_{u,t}$ the exposure policy, and $\Phi_{u,t}$ the composite feedback signal. All user windows use `window_size = 20`, and video tags are binned from `video_features_basic_pure.csv`. The benchmark keeps only users with at least 20 interactions in the healthy, bubble, and collapse segments, yielding the 367-user evaluation cohort used in the main text.

Operational score components

In KuaiRand, σ_P is a normalized deviation of the current watch-ratio window from the healthy watch-ratio baseline, σ_Φ is a stability term based on within-window watch-ratio dispersion, and σ_A is an operational surrogate based on normalized exposure-tag entropy. This last term should be read as a logged-data proxy for action reproducibility, not as a direct estimator of $e^{-H(\mathcal{A}|\mathcal{P})}$, because the benchmark has no replicated policy logs.

The proxy hierarchy matters: if direct action logs or replica logs exist, the Gaussian-style exact interpretation can be approximated; if not, the logged-data entropy surrogate is the intended operational choice. The logged benchmark is evidence for a useful proxy, not a proof that the surrogate equals the unobserved conditional entropy. KuaiRand is a proxy-level benchmark.

Effort proxy

The effort proxy is defined as the divergence between the current exposure distribution and the healthy random-exposure baseline distribution:

$$E_{u,t} := D_{\text{KL}} \left(\pi_{u,t}^{\text{current}} \parallel \pi_{u,t}^{\text{healthy}} \right).$$

This choice is directional, decomposable, and directly interpretable as the cost of keeping the observed trajectory close to a healthy baseline. In settings where only aggregate exposure counts are available, total variation distance may be used as a fallback, but KL is the preferred default. In the follow-up proxy sweep, the KL

proxy also outperformed the total-variation and exposure-Gini alternatives at the same healthy-only calibration, so it remains the default operational choice here. We use additive smoothing 10^{-6} in the KL computation.

Protocol

The benchmark unfolds in three phases.

1. **Healthy.** Exposure follows the random baseline, diversity is unconstrained, and effort remains low.
2. **Bubble.** Exposure becomes progressively concentrated, so that $E_{u,t}$ rises while standard coherence may remain artificially high.
3. **Collapse.** The concentrated exposure regime stops tracking the healthy baseline, generic detectors begin to fire, and the effort-corrected index should lead the baselines.

Evaluation

Evaluation is carried out sequentially within each user session and then batched across users with strict temporal splits. Thresholds for CI, CI^E , and EWMA-smoothed CI^E are calibrated on healthy windows only, using the 20th percentile of the healthy score distribution. The reference effort scale E_0 is the median healthy KL divergence. Reported metrics are:

- detection rate of the bubble/collapse phases,
- median detection delay,
- false positives in the healthy phase,
- sensitivity to λ in CI^E ,
- comparison against ADWIN, PageHinkley, KSWIN, NoDrift, and a smoothed CI^E baseline (EWMA).

The detection metric is user-level rather than AUC-based: a user is counted as detected for bubble or collapse if the detector issues at least one warning within a fixed four-window neighborhood before the corresponding annotated phase boundary. Reported uncertainty intervals are paired bootstrap intervals over users. All baseline detectors consume the same effort-adjusted scalar signal, $1 - CI^E$, so the comparison is not biased by detector-specific inputs. The baseline hyperparameters are fixed a priori: ADWIN $\delta = 0.03$; PageHinkley $\delta = 0.005$, threshold = 20, $\alpha = 0.9999$; KSWIN window = 30, stat = 10, $\alpha = 0.001$. When enabled, thresholds are calibrated on healthy windows only.

Controls

Required controls include shuffled labels, random phase boundaries, no- coercion ablations, and removal of auxiliary feedback signals. These controls keep the benchmark from collapsing into a hindsight-tuned logged-bandit exercise. We additionally report a clipped self-normalized tag-reweighting sanity check as a weak logged-data control, but we do not present that control as a causal IPS or counterfactual identification result because the repository lacks item-level propensities and replica policies.